Name:____________

GCSE Chemistry Unit 1 Higher Tier

Once you have completed a paper and have marked it, fill out the table below, using the grade boundaries provided to add your grade. Then list which questions you did well and which questions you struggled with. For the questions your struggled with, review why and at the reasons. The most common reasons are

- Not understanding basic concepts (e.g. simple definitions)

- Calculation errors (forgetting to square numbers, etc.)

- Not reading questions carefully

- Not answering the full question (describe vs. explain)

- Laboratory/practical setup mistakes

For these questions it is also useful add notes to the question to help you next time and take a photo of these questions. Use these then to guide further revision.

Year	Score /80	Grade	Good Questions	Bad Questions	Reasons
2017					
2018					
2019					
2022					
2023					

Grade boundaries - Raw Score

Year	E	D	C	B	A	A*	Max
2017	8	13	22	31	40	49	58
2018	9	12	18	29	41	53	65
2019	5	10	19	28	37	46	55
2022	5	7	14	25	37	49	61
2023	9	15	23	34	45	56	67

Surname	Centre Number	Candidate Number
Other Names		0

GCSE – NEW

3410UA0-1

S17-3410UA0-1

CHEMISTRY – Unit 1:
Chemical Substances, Reactions and Essential Resources

HIGHER TIER

FRIDAY, 16 JUNE 2017 – MORNING

1 hour 45 minutes

For Examiner's use only		
Question	**Maximum Mark**	**Mark Awarded**
1.	**9**	
2.	**11**	
3.	**8**	
4.	**6**	
5.	**11**	
6.	**12**	
7.	**8**	
8.	**9**	
9.	**6**	
Total	**80**	

3410UA01
01

ADDITIONAL MATERIALS

In addition to this paper you may require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **9** is a quality of extended response (QER) question where your writing skills will be assessed.

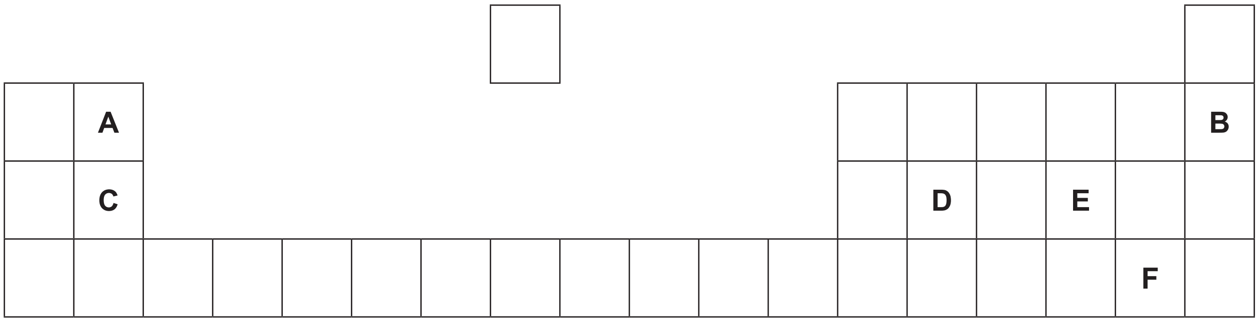
The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.

Examiner only

Answer **all** *questions*.

1. *(a)* The following diagram shows an outline of part of the Periodic Table.

The letters shown are NOT the chemical symbols of the elements.

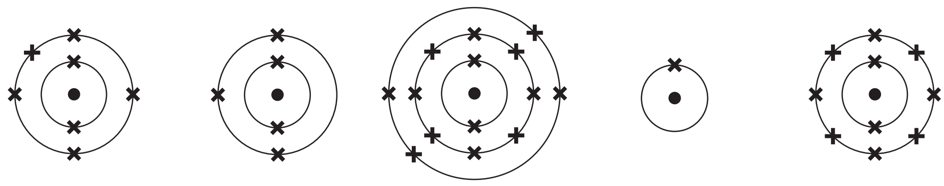


Choose **letters** from the diagram to complete the table below. [5]

	Letter(s)
Two elements in the same group	.... and ....
The element which has 12 protons in its nucleus	....
The element with a full outer shell of electrons	....
The element in Group 2 and Period 2	....
The element with the electronic structure 2,8,4	....

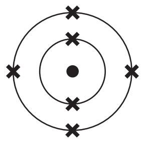
Examiner only

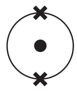
(b) Diagrams **1-5** show the electronic structure of five elements in the Periodic Table.



1 2 3 4 5

Give the **number** of the diagram which shows the electronic structure of the element which lies

(i) directly **below**  in the Periodic Table, ... [1]

(ii) to the **left** of  in the Periodic Table. .... [1]

3410NA01 03

(c) Nitrogen has two stable isotopes – nitrogen-14 and nitrogen-15.

Describe how these isotopes are similar to one another and how they are different. [2]

....

....

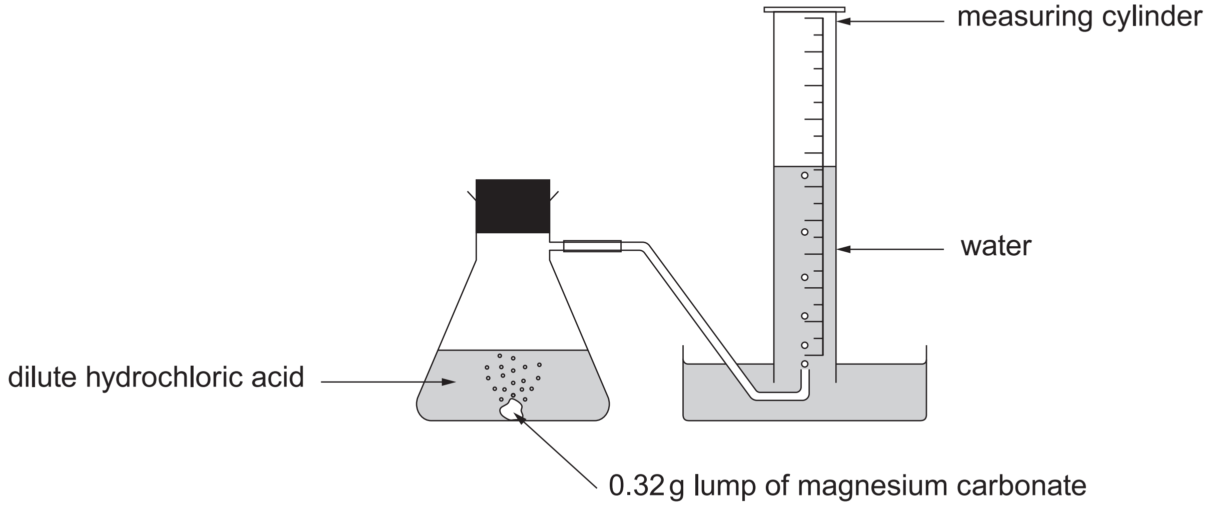
....

9

Examiner only

2. A student carried out an experiment to investigate the speed of the reaction between a **lump** of magnesium carbonate of mass 0.32 g and excess dilute hydrochloric acid at 20 °C.

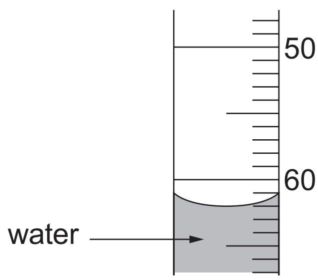
Magnesium carbonate reacts with dilute hydrochloric acid forming carbon dioxide gas. The total volume of carbon dioxide formed was recorded every 5 minutes for 40 minutes.



The results are shown below. The result for 15 minutes is missing.

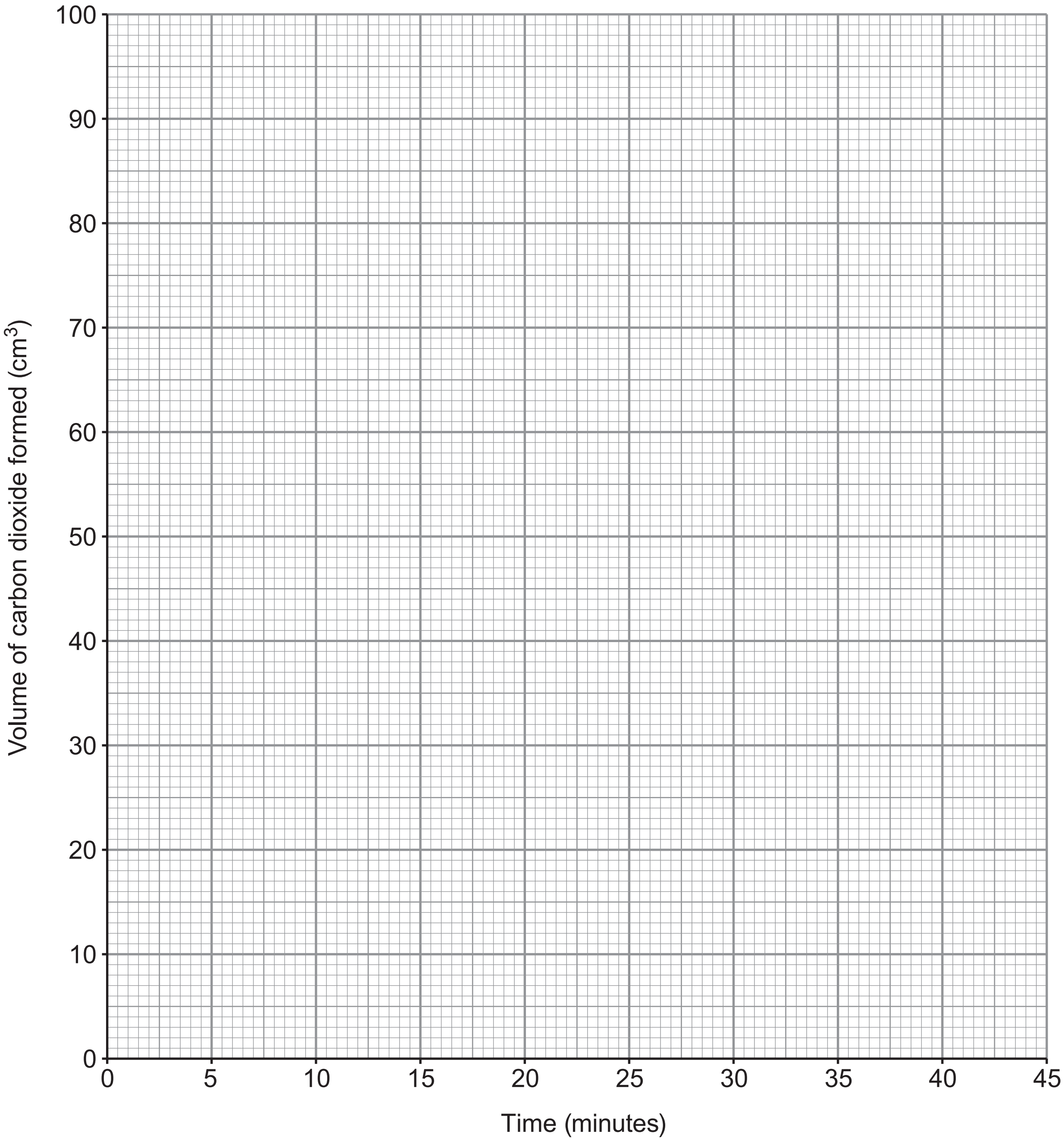
Time (minutes)	0	5	10	15	20	25	30	35	40
Volume of carbon dioxide formed (cm^3)	0	20	41		79	83	90	90	90

(a) Use the diagram below to find the volume of carbon dioxide gas formed after 15 minutes. [1]



volume of carbon dioxide = ... cm^3

(b) Plot the results from the table, including your answer to part *(a)*, on the grid below. Draw a suitable line and **label this graph X**. [3]



(c) Sketch the graph you would expect if the experiment were repeated using 0.32 g of magnesium carbonate **powder** instead of the lump of magnesium carbonate.
Label this graph Y. [2]

Examiner only

(d) State and explain, using particle theory, the effect of **increasing** the concentration of the hydrochloric acid. [3]

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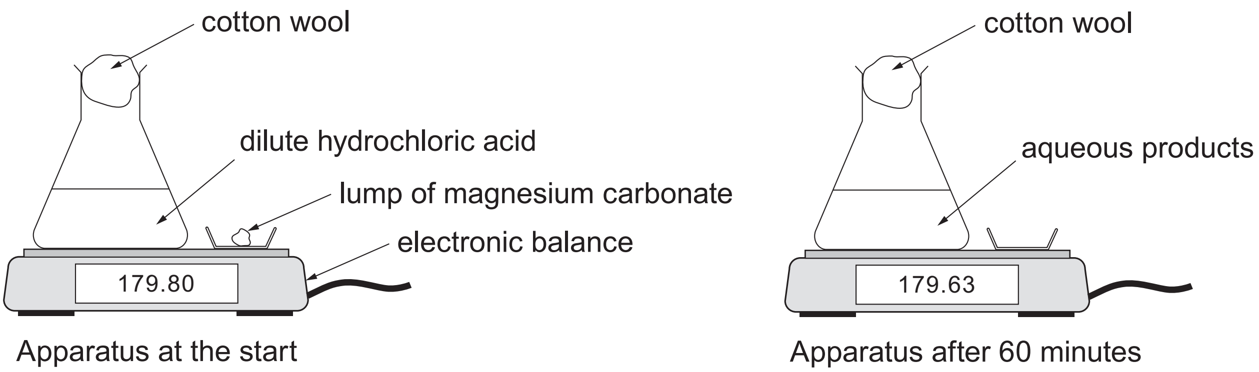
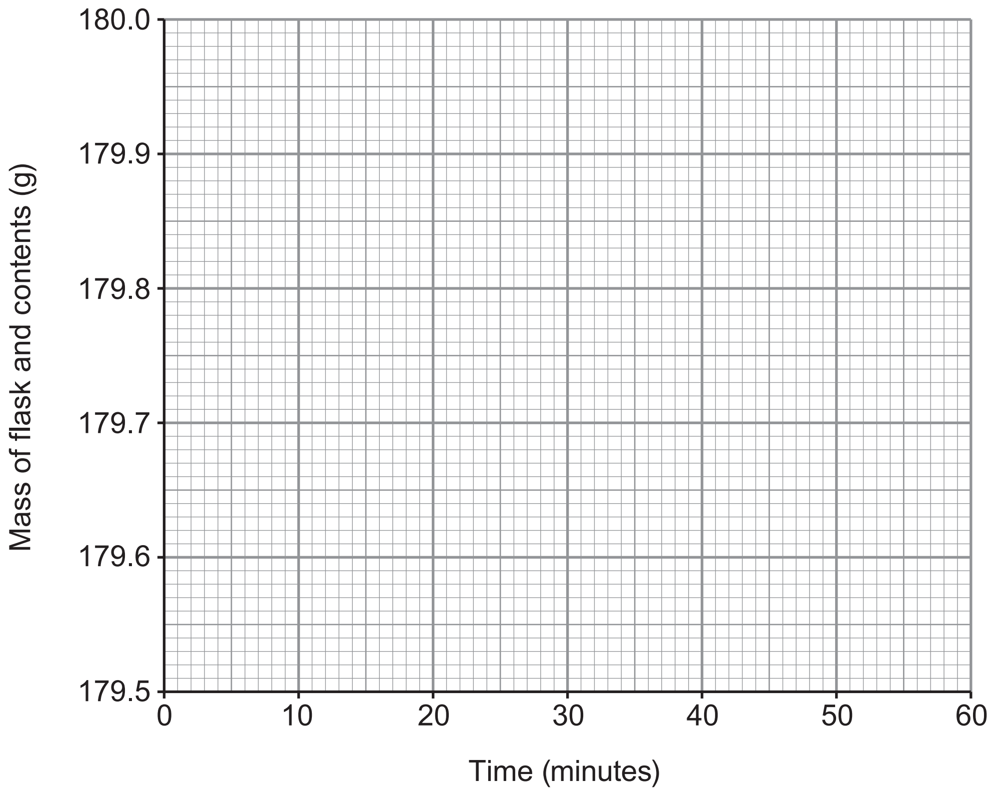
Examiner only

(e) The student investigated the same reaction using a different apparatus. A lump of magnesium carbonate was added to excess dilute hydrochloric acid at 20 °C.



The change in mass was recorded for 60 minutes and displayed as a graph on a computer screen. The reaction took 40 minutes to complete.

Sketch the graph you would expect to see. [2]



3410NA01 07

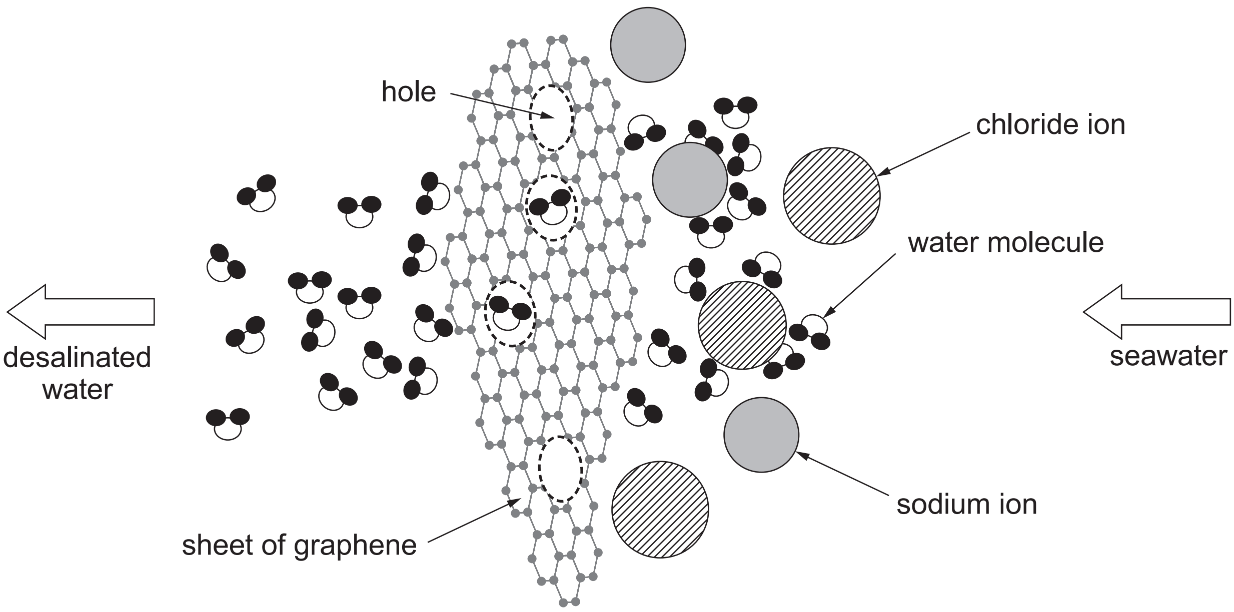
11

3. *(a)* One out of six people today do not have access to safe fresh water. Estimates suggest fresh water supplies will be a major problem for half the countries of the world by 2025 and by 2050 about 75 % of the world's population will experience a serious scarcity of the resource.

As the world's population increases not only will the demand for drinking water and water for sanitation increase, but the demand for water to produce more food will increase. As climate change makes rainfall less predictable and droughts more common, a growing number of countries are turning to desalination.

One method of desalination, called reverse osmosis, uses a thick membrane to filter salt from seawater. But this system requires an extremely high pressure to force water through the membrane and therefore uses a lot of energy.

Researchers in the USA have come up with a new approach using a different kind of filtration material - sheets of graphene - a one-atom-thick form of the element carbon. This new material can be far more efficient and possibly less expensive than existing desalination methods. Sheets of graphene are perforated with precisely-sized holes which only let the water molecules through.



Examiner only

(i) Tick (✓) the box next to the **main** reason why countries are building desalination plants. [1]

water is becoming scarce ☐

each person is drinking more water ☐

climate change affects the availability of drinking water ☐

(ii) Tick (✓) the statement which **best** explains how graphene sheets desalinate water. [1]

water molecules contain atoms and not ions ☐

sheets contain holes which are bigger than ions ☐

sheets contain holes which are bigger than water molecules but smaller than sodium and chloride ions ☐

sheets contain holes which let molecules through but not ions ☐

molecules are smaller than ions ☐

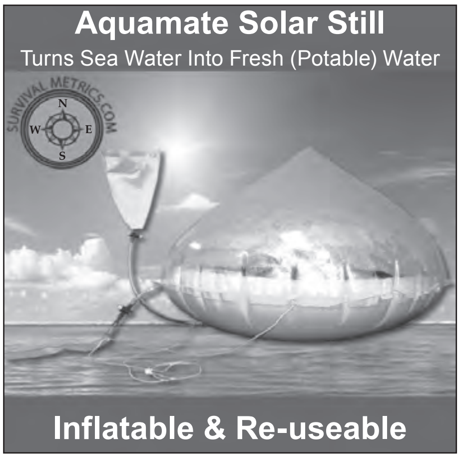
(b) The total volume of water on the Earth is 1.4×10^9 km^3, 97 % of which is seawater. Most of the remaining 3 % is bound up in ice at the poles, leaving 0.3 % available as fresh water.

Calculate the volume of the Earth's available fresh water. Give your answer in standard form. [2]

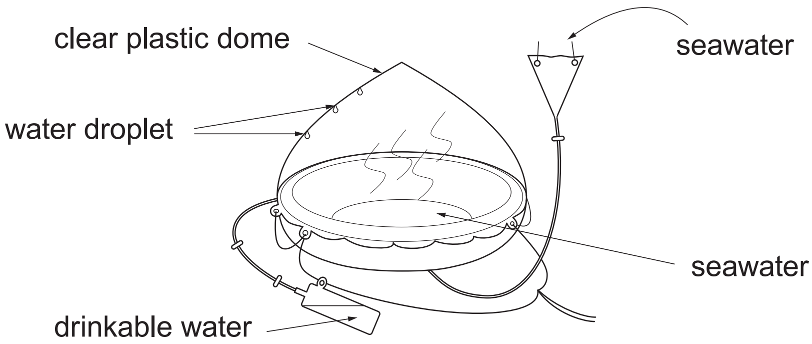
volume of available fresh water = ... km^3

Examiner only

(c) One piece of survival equipment found on life rafts is a ‘solar still’. Solar stills are used to remove soluble salts from seawater to form drinkable water.



Source: *www.survivalmetrics.com*



State and explain the physical changes that occur in the formation of drinkable water from seawater. Include the name given to the overall process taking place. [4]

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8

Examiner only

4. *(a)* Limestone quarrying is an important business in the UK. Quarrying creates jobs in areas where there are often limited opportunities. There is a huge demand for the products of quarrying, such as building stone and cement. Good roads and rail links are needed for transporting the products of quarrying. Thousands of people are employed in quarrying and related industries.

Many people argue against the opening of a quarry in their area despite all the benefits of limestone quarrying. Explain **two** reasons local people might use in an attempt to stop the opening of a new quarry in their area. [2]

....

....

....

....

....

(b) When limestone is heated in a lime kiln thermal decomposition takes place.

Write a balanced **symbol** equation for this reaction. [1]

....

3410NA01 11

(c) Calcium silicide is a compound containing calcium and silicon only. A sample of calcium silicide was found to contain 2.0 g of calcium and 2.8 g of silicon.

Calculate the simplest formula for this compound. You **must** show your working. [3]

Simplest formula ....

6

Examiner only

5. *(a)* Gwyn carried out an experiment to find the solubility of potassium nitrate at room temperature, 25 °C. The method he used is shown in stages **1**-**3**.

Stage **1**

He added solid potassium nitrate to 50 cm^3 of water until a saturated solution was formed.

⇩

Stage **2**

He pipetted 25 cm^3 of the saturated solution into a pre-weighed evaporating basin. He put the basin and the solution into a warm oven.

⇩

Stage **3**

He removed the basin from the oven and allowed it to cool. He weighed the basin and crystals after 2 hours.

His results are shown below.

Mass of evaporating basin	= 42.6 g
Mass of evaporating basin and potassium nitrate crystals	= 54.2 g

(i) State what is meant by a *saturated* solution. [1]

....

....

(ii) Use Gwyn's results to calculate the solubility of potassium nitrate in g / 100 g of water. [2]

solubility = ... g / 100 g of water

Examiner only

(iii) Gwyn's experimental value for the solubility of potassium nitrate is much bigger than expected. Suggest how Gwyn should adapt stage **3** of his method in order to get a more accurate value. Explain your reasoning. [3]

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(iv) Gwyn was asked to find out if sodium chloride is more soluble in water than potassium nitrate. State which variable Gwyn **must keep the same** in his experiment in order for him to be able to compare the two solubilities. [1]

....

(b) (i) The solubility of potassium bromide at 20 °C is 64 g / 100 g of water. Calculate the mass of solid that forms when a solution containing 43.9 g of potassium bromide in 50 g of water is cooled to 20 °C. [2]

mass = ... g

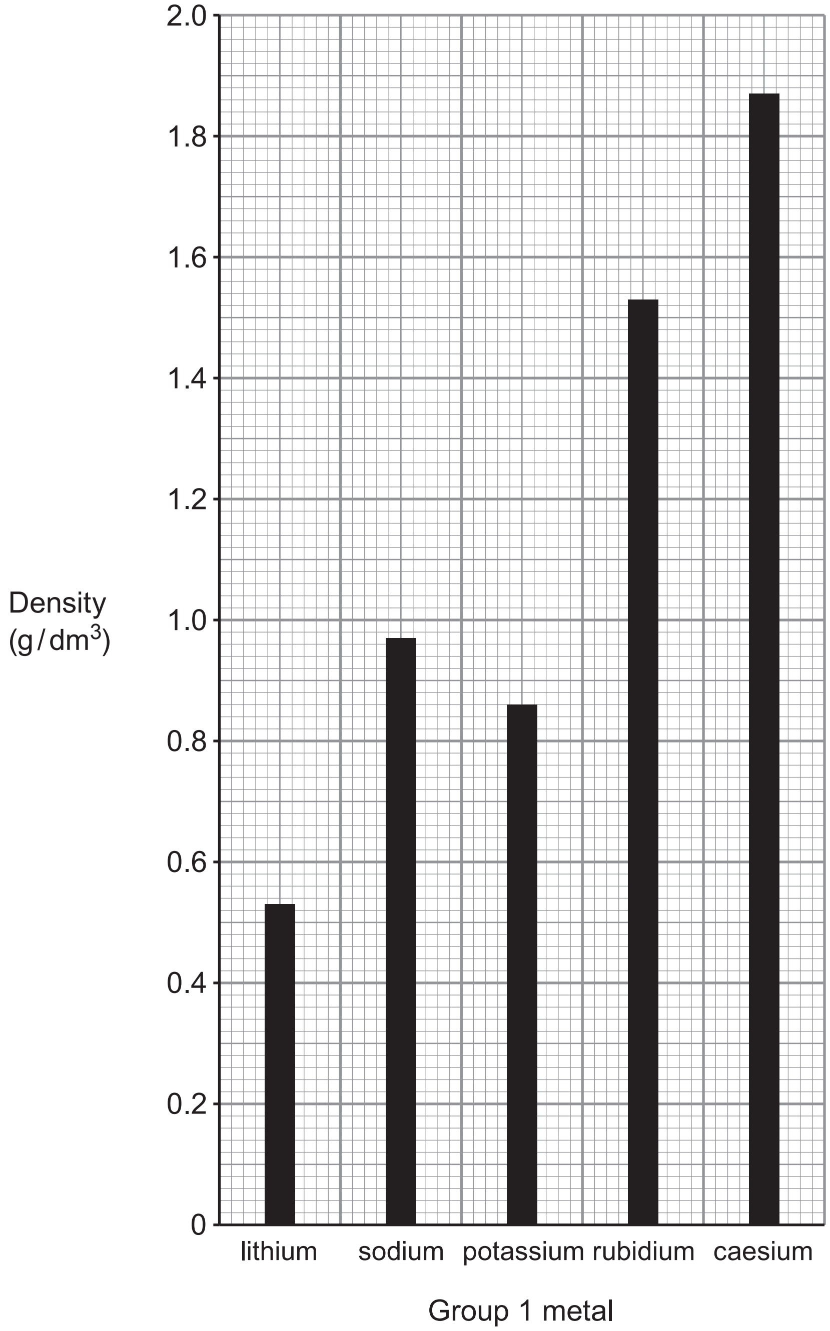
(ii) Calculate the number of moles of potassium bromide, KBr, in 64 g. Give your answer to **two** significant figures. [2]

number of moles = ... mol

11

Examiner only

6. *(a)* The bar graph shows the densities of five Group 1 metals.



Draw a **straight** line on the bar graph showing the general trend in density of Group 1 metals. How does your trend line suggest that potassium and **not** sodium is the anomalous density value? [2]

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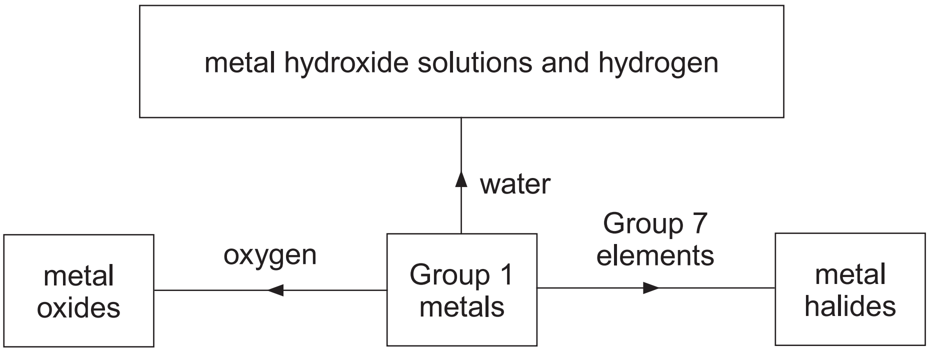
Examiner only

(b) The flow chart shows some reactions of Group 1 metals.



(i) State and explain **one** similarity and **one** difference you would see when lithium and potassium are added separately to a trough of cold water. [4]

Similarity ....

....

Difference ....

....

(ii) Write a balanced **symbol** equation to show the reaction between lithium and oxygen. [3]

....

(iii)

Group 1	**Group 7**
lithium sodium potassium	chlorine bromine iodine

The boxes show some of the elements in Group 1 and Group 7. Explain, in terms of electronic structure, why potassium and chlorine would react the most violently. [3]

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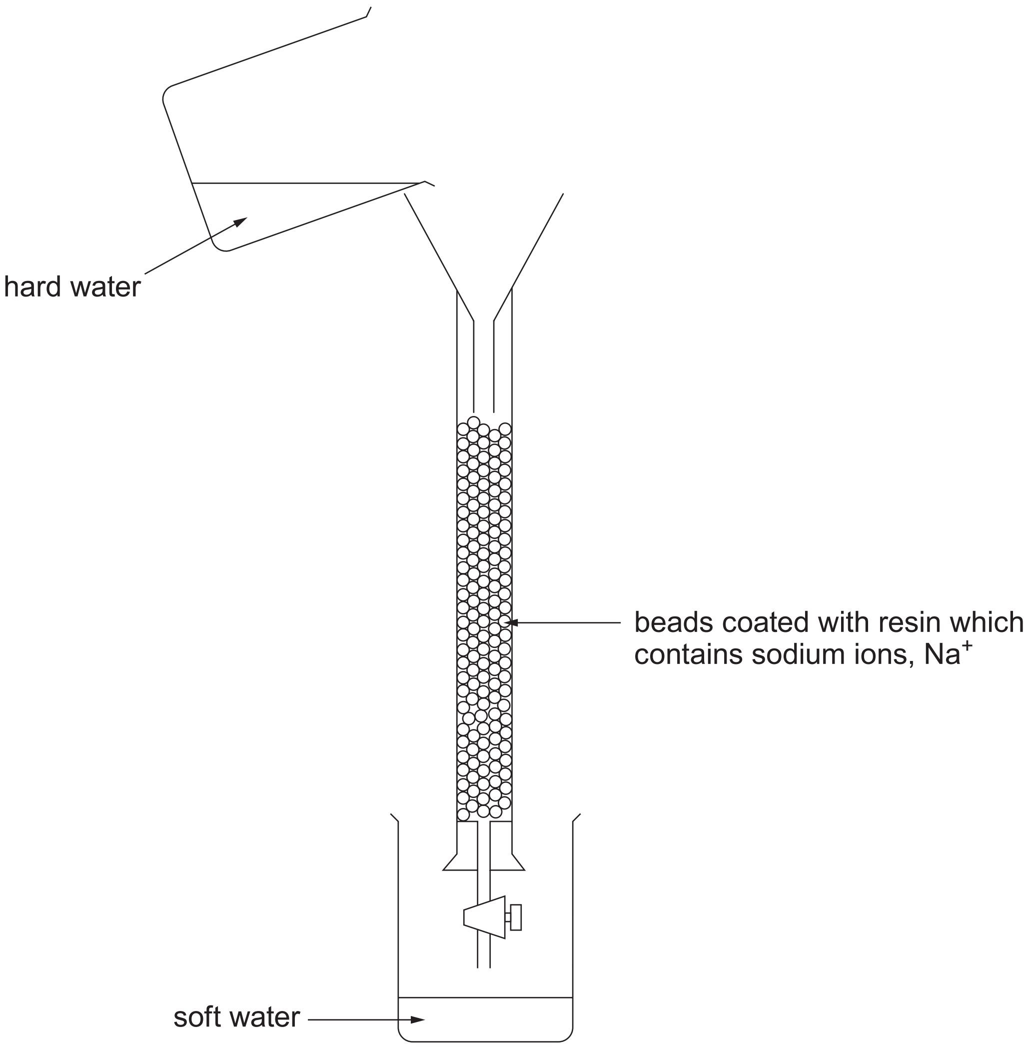
12

Examiner only

7. *(a)* The diagram shows an ion exchange column used for softening permanent hard water.



(i) Explain how ion exchange works. [2]

....

....

....

(ii) Explain why the ion exchange resin will stop working after continued use. Name a solution which can be passed through the resin to regenerate it. [2]

....

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Examiner only

(b) Describe and explain a method to test whether a water sample is soft water, temporary hard water or permanent hard water. [4]

(You do **not** need to include reference to fair testing in your answer)

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8

Examiner only

8. *(a)* (i) The table below shows results of tests carried out on three white Group 1 compounds, **A**, **B** and **C**.

The flame tests were carried out on solids and the silver nitrate tests on solutions of the compounds.

Test	Compound **A**	Compound **B**	Compound **C**
Flame test	red	yellow	lilac
Add silver nitrate solution	white precipitate	cream precipitate	yellow precipitate

Use the information in the table to identify the compounds. [2]

Compound **A** ....

Compound **B** ....

Compound **C** ....

(ii) The symbol equation represents the reaction occurring between solutions of silver nitrate and magnesium chloride.

$$2AgNO_3(aq) + MgCl_2(aq) \longrightarrow Mg(NO_3)_2(aq) + 2AgCl(s)$$

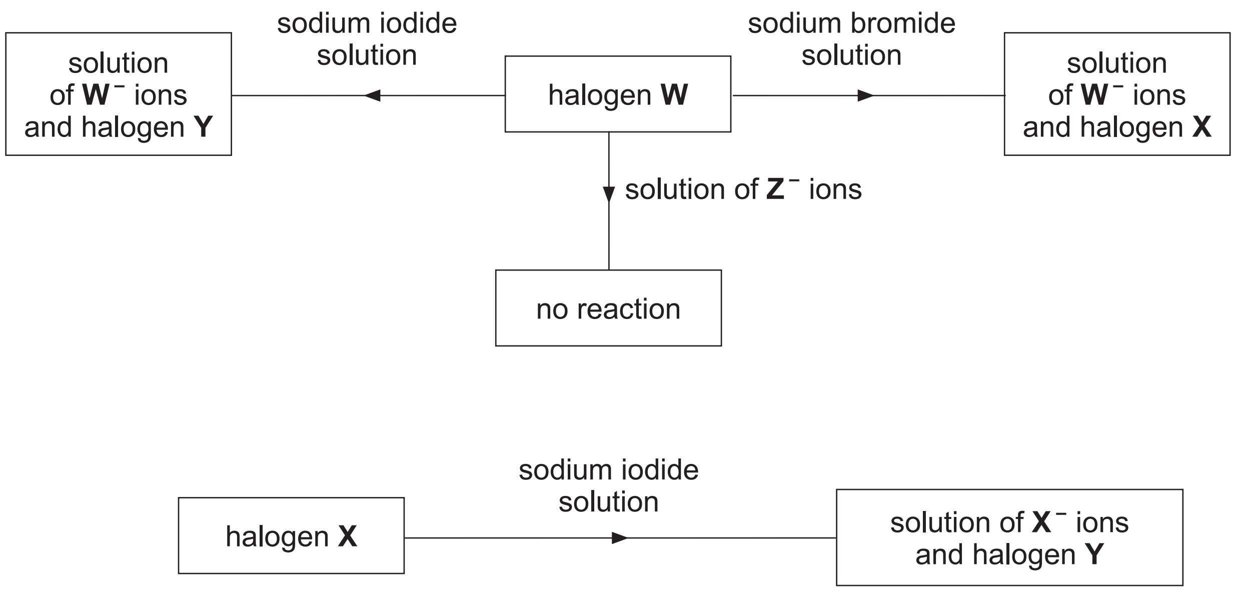
Write the **ionic** equation for the reaction. Include state symbols in your answer. [3]

....

Examiner only

(b) **W**, **X**, **Y** and **Z** represent the halogens fluorine, chlorine, bromine and iodine, but not necessarily in that order.

The diagrams below show some reactions of halogens **W** and **X**.



Use the information to identify halogens **W**, **X**, **Y** and **Z**. [2]

W ....

X ....

Y ....

Z ....

(c) Iron wool burns in fluorine to give iron(III) fluoride.

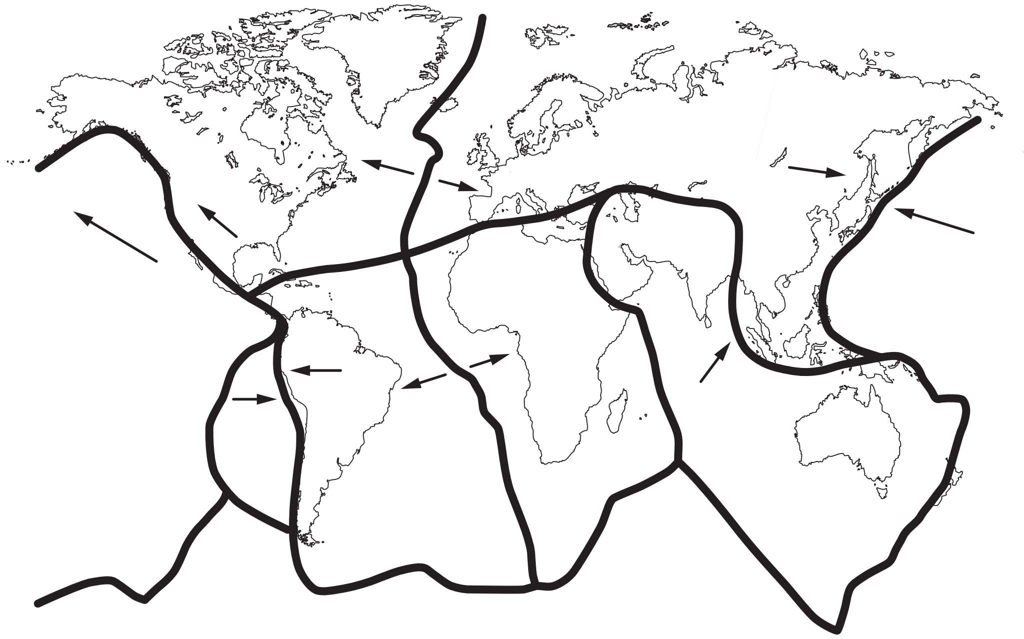
Write a balanced **symbol** equation for this reaction. [2]

....

9

Examiner only

9. The diagram shows the movement of the major tectonic plates.



Explain what happens at destructive and constructive plate boundaries.
Diagrams may be included in your answer. [6 QER]

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6

END OF PAPER

Examiner only

For continuation only.

FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	**Formula**	**Name**	**Formula**
aluminium	Al^{3+}	bromide	Br^{-}
ammonium	NH_4^{+}	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^{-}
calcium	Ca^{2+}	fluoride	F^{-}
copper(II)	Cu^{2+}	hydroxide	OH^{-}
hydrogen	H^{+}	iodide	I^{-}
iron(II)	Fe^{2+}	nitrate	NO_3^{-}
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^{+}	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^{+}		
silver	Ag^{+}		
sodium	Na^{+}		
zinc	Zn^{2+}		

THE PERIODIC TABLE

Group

1	2											3	4	5	6	7	0
						1 H Hydrogen 1											4 He Helium 2
7 Li Lithium 3	9 Be Beryllium 4											11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10
23 Na Sodium 11	24 Mg Magnesium 12											27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
86 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86
223 Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89															

Key

A_r — relative atomic mass
Symbol
Name
Z — atomic number

Surname	Centre Number	Candidate Number
Other Names		0

GCSE – NEW

3410UA0-1

S18-3410UA0-1

CHEMISTRY – Unit 1:
Chemical Substances, Reactions and Essential Resources

HIGHER TIER

WEDNESDAY, 13 JUNE 2018 – MORNING

1 hour 45 minutes

For Examiner's use only		
Question	**Maximum Mark**	**Mark Awarded**
1.	**6**	
2.	**9**	
3.	**5**	
4.	**9**	
5.	**11**	
6.	**6**	
7.	**10**	
8.	**10**	
9.	**8**	
10.	**6**	
Total	**80**	

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ADDITIONAL MATERIALS

In addition to this examination paper you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **6** is a quality of extended response (QER) question where your writing skills will be assessed.

The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.

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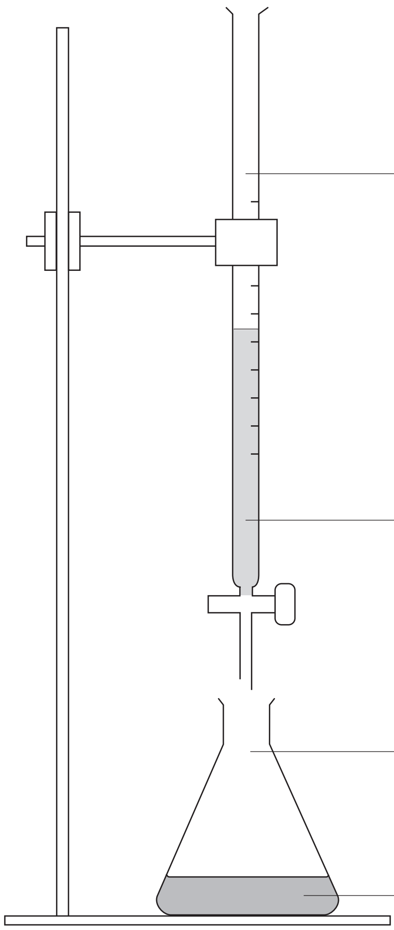
Answer **all** *questions.*

1. Water samples **A**, **B**, **C** and **D** were tested for hardness using the apparatus shown.



Soap solution was added 1 cm^3 at a time to each sample and the volume required to produce a permanent lather on shaking was recorded. Each sample was tested before and after boiling. The results are shown in the table.

Water sample	Volume of soap solution required (cm^3)	
	Before boiling	After boiling
A	1	1
B	10	10
C	15	1
D	15	8

Examiner only

(a) (i) State which water sample contains **only** temporary hardness. Explain your answer. [2]

Water sample ....

Explanation ....

....

....

(ii) Give **one** similarity in the composition of temporary and permanent hard water. [1]

....

....

(b) Discuss the benefits and drawbacks of living in a hard water area. [3]

....

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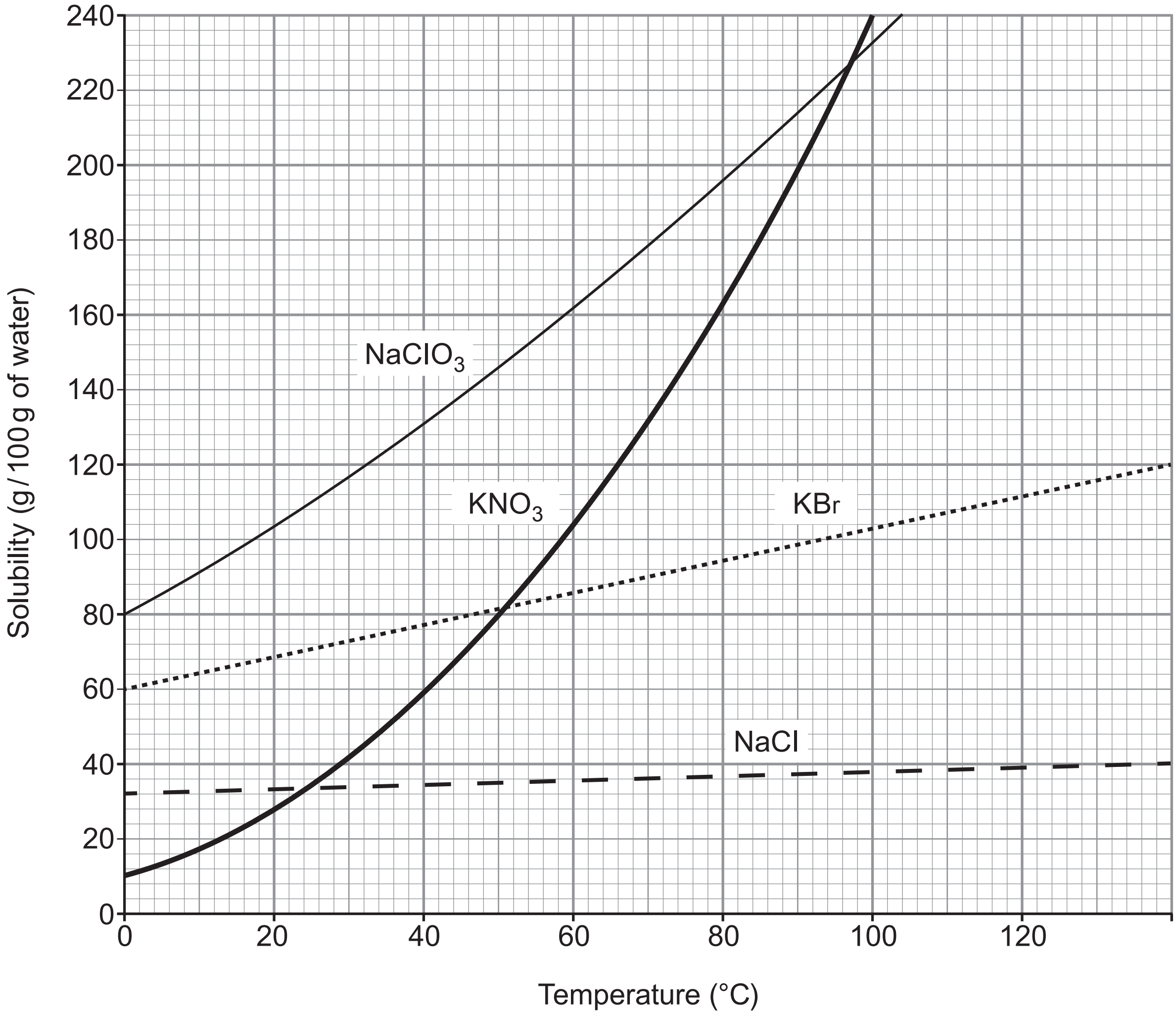
6

2. The grid below shows the solubility curves for four ionic compounds.



$NaClO_3$	sodium chlorate
KNO_3	potassium nitrate
KBr	potassium bromide
NaCl	sodium chloride

Examiner only

(a) (i) Give the temperature at which the solubility of potassium nitrate and potassium bromide is the same. [1]

.... °C

(ii) Calculate the mass of solid potassium nitrate that would form if a saturated solution in 200 g of water were cooled from 100 °C to 20 °C. [3]

Mass = ... g

(iii) Suggest why a student may be surprised at the temperature range shown on the solubility curves. [1]

....

....

(b) (i) Give the symbols of the **ions** of Group 1 elements present in the compounds shown on the grid. [1]

....

(ii) Explain how these ions are formed from their atoms. [2]

....

....

(c) Potassium nitrate reacts with aluminium hydroxide to produce aluminium nitrate and potassium hydroxide.

Balance the symbol equation for the reaction taking place. [1]

$$\square\ KNO_3 + Al(OH)_3 \longrightarrow Al(NO_3)_3 + \square\ KOH$$

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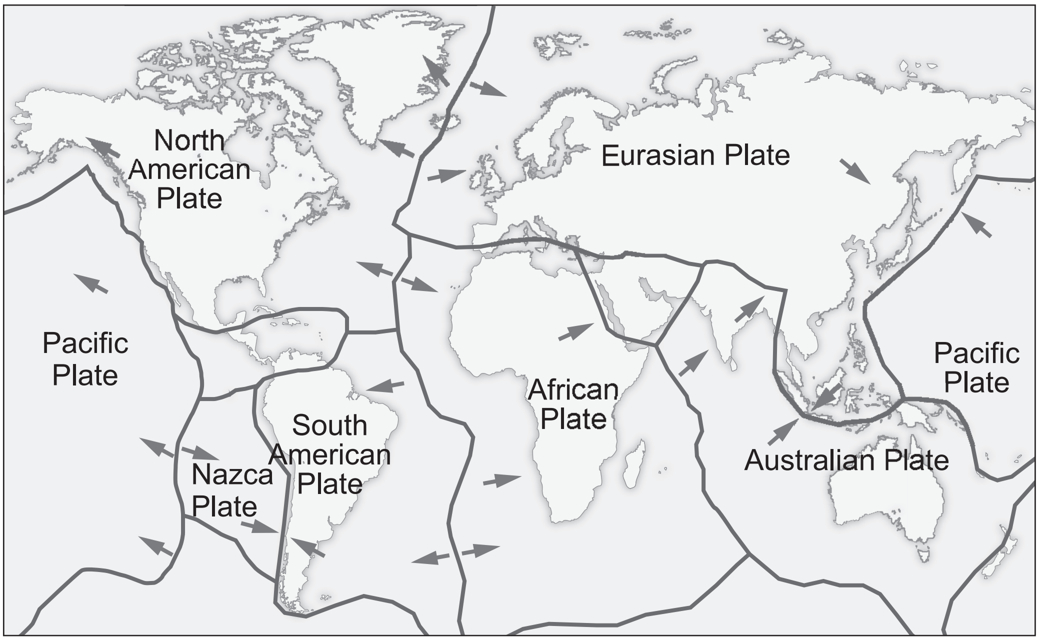
9

Examiner only

3. The following diagram shows some of the Earth's tectonic plates and the direction in which they move.



(a) The boundary between the Nazca and South American plates is a destructive plate boundary. Describe what happens at a destructive boundary. [2]

....

....

....

(b) Draw a cross (**×**) on the diagram to show a constructive plate boundary. Describe what happens at this boundary. [2]

....

....

....

(c) State **one** effect of plates sliding past each other. [1]

....

5

Examiner only

4. *(a)* Dilute hydrochloric acid reacts with sodium thiosulfate to make the products shown in the equation.

$$\boxed{\ ?\ }(aq) + 2HCl(aq) \longrightarrow 2NaCl(aq) + SO_2(g) + S(s) + H_2O(l)$$

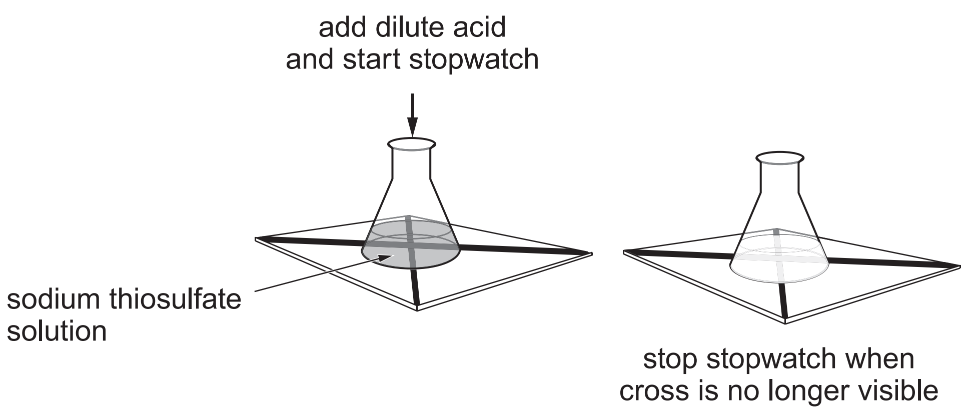
(i) Use the equation to work out the formula of sodium thiosulfate. [1]

Formula ....

(ii) The symbol (aq) in the equation tells us that the substances are aqueous. What is meant by this? [1]

....

(iii) The rate of this reaction can be studied as shown in the diagram.



Use information **from the equation** to explain why the cross disappears. [2]

....

....

....

Examiner only

(b) A student studied the effect of temperature on the rate of this reaction. He obtained the following results.

Temperature (°C)	Time taken for cross to disappear (s)			
	1	2	3	Mean
15	130	128	129	129
30	53	53	53	53
45	21	29	23	24.3
60	7	7	6	6.7

(i) Another student said that one of the mean values was incorrect.
Identify the incorrect mean. Give your reasoning. [2]

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(ii) State what conclusion can be drawn about the effect of temperature on the rate of this reaction. Explain your conclusion using particle theory. [3]

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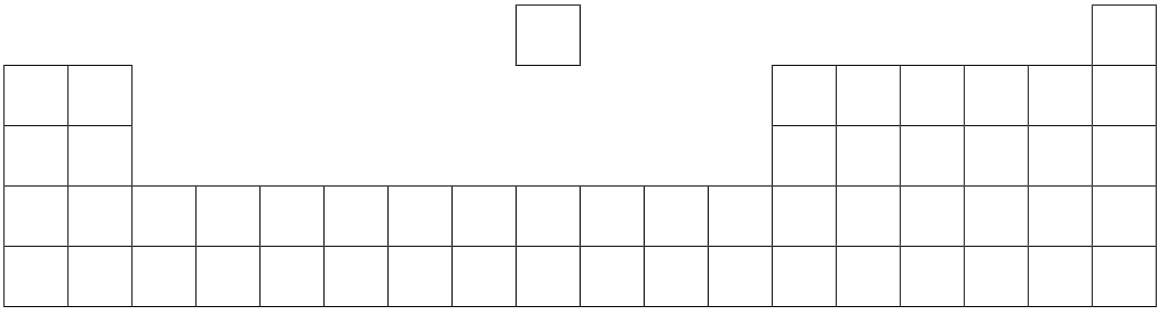
9

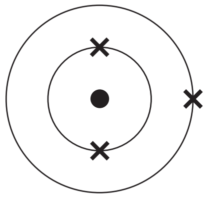
Examiner only

5. The following diagram is an outline of the Periodic Table.



(a) Write letters **A**, **B** and **C** on the **diagram** in the positions of the elements that fit the following descriptions. [3]

A the element with the electronic structure



B the element in Group 2 and Period 4

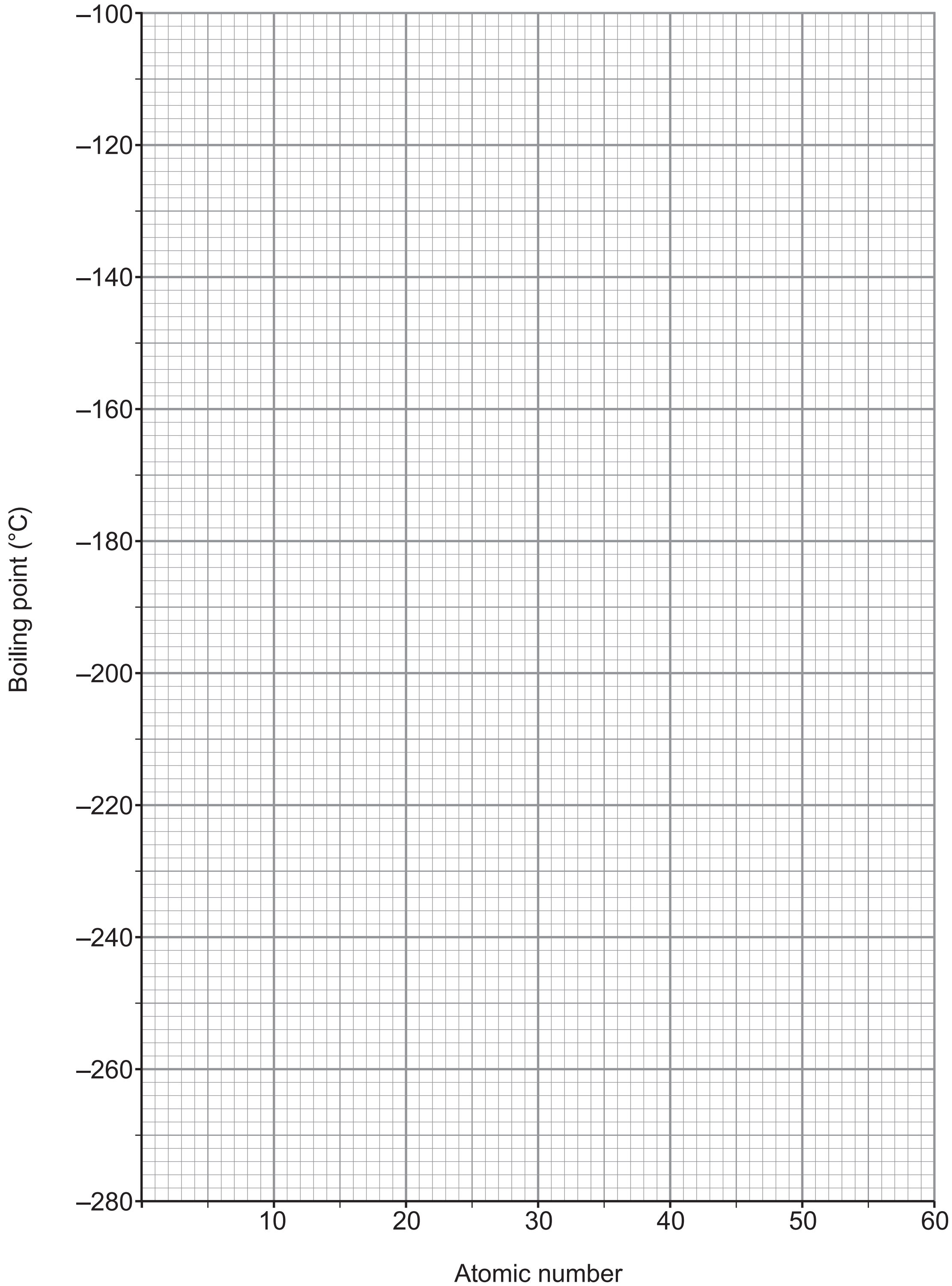
C an element that shows both metallic and non-metallic properties

(b) The following table shows the atomic numbers and boiling points of the inert gases.

Inert gas	helium	neon	argon	krypton	xenon
Atomic number	2	10	18	36	54
Boiling point (°C)	−269	−246	−186	−153	−108

(i) Plot this data on the grid opposite. Draw a suitable line. [3]

Examiner only



(ii) Describe the trend in boiling point shown on the graph. [1]

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Examiner only

(c) The following table shows the boiling points of the inert gases in both °C and K.

Inert gas	helium	neon	argon	krypton	xenon
Boiling point (°C)	−269	−246	−186	−153	−108
Boiling point (K)	4	27	....	120	165

Use the information in the table to calculate the boiling point of argon in K. [2]

Boiling point = ... K

(d) Give **one** use of argon. Explain in terms of electronic structure why it is used for this purpose. [2]

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11

Examiner only

6. Limestone is an important raw material. It can be used as a building material or converted into quicklime and slaked lime.

Describe and explain the sequence of reactions carried out in the laboratory to convert limestone into slaked lime. [6 QER]

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6

Examiner only

7. *(a)* Group 7 elements are known as the halogens. The following table shows the observations made when the first three members of the group react with hydrogen.

Halogen	Observations
fluorine	explodes in cold and dark
chlorine	explodes in sunlight
bromine	small explosion when ignited with a flame

(i) Use your knowledge of electronic structure to explain why all the halogens react in a similar way and why they react more slowly on going down the group. [3]

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(ii) Hydrogen fluoride is highly corrosive and can be used to etch glass which is mainly silicon dioxide.

Balance the symbol equation for the reaction between hydrogen fluoride and silicon dioxide. [1]

$$\boxed{}\ HF + SiO_2 \longrightarrow SiF_4 + \boxed{}\ H_2O$$

(iii) Calcium fluoride reacts with sulfuric acid, H_2SO_4, to produce calcium sulfate and hydrogen fluoride. Give the **symbol** equation for the reaction. [3]

....

Examiner only

(b) Chlorine reacts with aluminium to produce aluminium chloride.

A sample of aluminium chloride of mass 26.70 g was found to contain 5.45 g of aluminium. Calculate the simplest formula of this chloride of aluminium.

You **must** show your working. [3]

$A_r(\text{Al}) = 27 \qquad A_r(\text{Cl}) = 35.5$

Formula ....

10

Examiner only

8. Sodium is extracted from sodium chloride.

(a) The overall reaction taking place is shown in the equation below.

$$2NaCl \longrightarrow 2Na + Cl_2$$

(i) When carrying out the reaction 120 kg of sodium chloride was found to produce 38.05 kg of sodium.

Calculate the maximum possible mass of sodium that could be produced and use this figure to calculate the percentage yield of this reaction. [4]

$A_r(Na) = 23$ $A_r(Cl) = 35.5$

Maximum possible mass = ... kg

Percentage yield = ... %

(ii) Suggest a reason why the yield is less than 100 %. [1]

....

....

(iii) Suggest why this reaction must be carried out under dry conditions. [1]

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Examiner only

(b) A sample of lithium is found to contain two isotopes.

Isotope	Percentage present in sample (%)
lithium-6	7.59
lithium-7	92.41

(i) Calculate the relative atomic mass (A_r) of lithium. Give your answer to **three** significant figures. [3]

$$A_r = \frac{(\text{isotope 1 mass} \times \text{abundance}) + (\text{isotope 2 mass} \times \text{abundance})}{100}$$

A_r = ...

(ii) State the difference between the atomic structures of lithium-6 and lithium-7. [1]

....

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10

Examiner only

9. The following table shows the decomposition temperatures of Group 2 metal carbonates and nitrates.

Metal	Temperature at which the carbonate decomposes (°C)	Temperature at which the nitrate decomposes (°C)
magnesium	117	89
calcium	178	561
strontium	235	570
barium	267	700

(a) Describe the trends in the stabilities of the Group 2 carbonates and nitrates. [3]

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(b) When a carbonate decomposes it produces carbon dioxide gas. Describe an experiment that could be carried out to show that carbon dioxide gas is produced during the reaction. [2]

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(c) When calcium nitrate decomposes it forms calcium oxide, oxygen and nitrogen dioxide, NO_2.

Write a **symbol** equation for the reaction. [3]

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8

10. Fluorine exists naturally as the fluoride ion. It is found in soil, water, foods and several minerals, such as fluorapatite and fluorite.

Fluoride ion concentration in seawater averages 1.3 ppm (parts per million). In fresh water, the natural range is typically between 0.01 and 0.3 ppm. In some parts of the world, fresh water contains fluoride ion levels which are dangerous and can lead to health problems.

In the early 1930s, scientists found that people who were brought up in areas with naturally fluoridated water had up to two-thirds fewer cavities compared to those who lived in areas where the water was not fluoridated. Several studies since then have repeatedly shown that when fluoride is added to people's drinking water in areas where natural levels are low, tooth decay decreases.

However, many European countries which do not fluoridate their water do not have a higher incidence of dental decay than countries which do so. It was also found that in Germany and Finland, decay rates either remained stable or continued in their downward trend after they stopped adding fluoride to their drinking water.

Figure 1 shows data about the effect of fluoridation of drinking water on the mean number of decayed, missing and filled teeth (DMFT) and the amount of fluorosis seen.

Figure 2 shows the change in mean DMFT in three regions of Australia over a four year period.

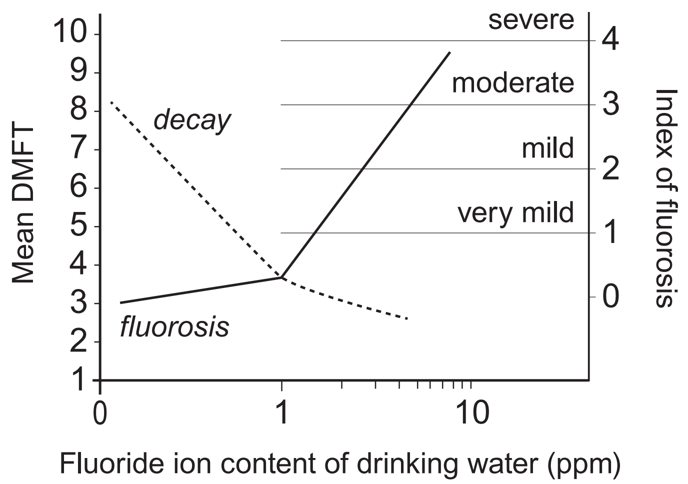

Figure 1

Examiner only

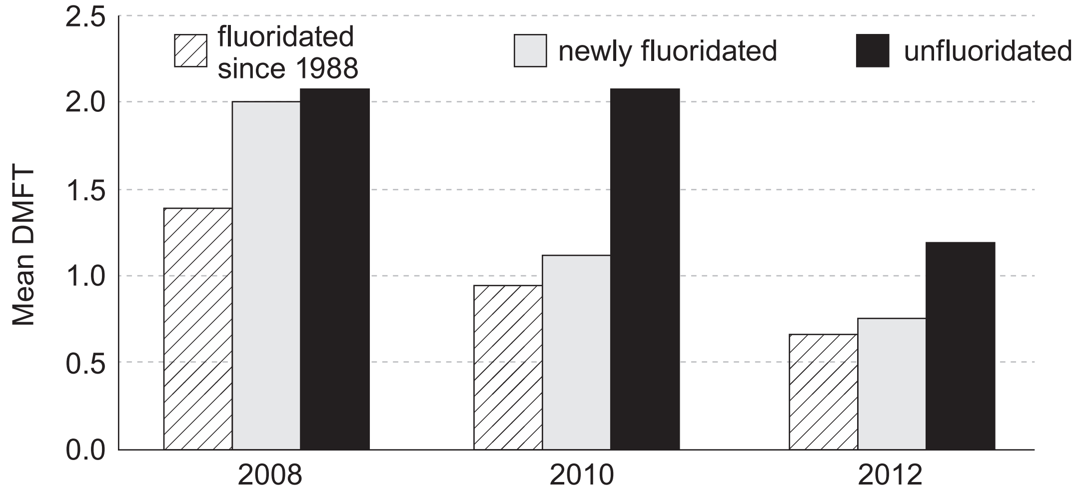

Figure 2

(a) Describe the effects of adding varying concentrations of fluoride ions to drinking water. [3]

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(b) Tick (✓) the statement below that best describes why Germany stopped fluoridating its water supplies. [1]

Ten years after stopping adding fluoride there was no increase in tooth decay ☐

They found that adding fluoride caused fluorosis ☐

Natural water supplies already contain fluoride in a high concentration ☐

Studies showed that areas with no fluoridation did not have higher levels of decay than areas that did fluoridate ☐

Examiner only

(c) Tick (✓) the box which gives **one** definite conclusion that can be drawn using **only** the data in **Figure 2**. [1]

Fluoridation has no effect on levels of decay	☐
People have reduced their intake of sugary foods over this period	☐
More than one factor affects levels of decay	☐
Fluoridation is the main cause of falling levels of decay	☐

(d) 'Mass medication' is an argument often given to oppose fluoridation of water supplies. Explain what is meant by the term *mass medication*. [1]

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6

END OF PAPER

FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	**Formula**	**Name**	**Formula**
aluminium	Al^{3+}	bromide	Br^{-}
ammonium	NH_4^{+}	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^{-}
calcium	Ca^{2+}	fluoride	F^{-}
copper(II)	Cu^{2+}	hydroxide	OH^{-}
hydrogen	H^{+}	iodide	I^{-}
iron(II)	Fe^{2+}	nitrate	NO_3^{-}
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^{+}	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^{+}		
silver	Ag^{+}		
sodium	Na^{+}		
zinc	Zn^{2+}		

THE PERIODIC TABLE

Group

1	2											3	4	5	6	7	0
						1 H Hydrogen 1											4 He Helium 2
7 Li Lithium 3	9 Be Beryllium 4											11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10
23 Na Sodium 11	24 Mg Magnesium 12											27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
86 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86
223 Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89															

Key

A_r	relative atomic mass
Symbol	
Name	
Z	atomic number

Surname	Centre Number	Candidate Number
Other Names		0

GCSE

3410UA0-1

S19-3410UA0-1

WEDNESDAY, 12 JUNE 2019 – MORNING

CHEMISTRY – Unit 1:
Chemical Substances, Reactions and Essential Resources

HIGHER TIER

1 hour 45 minutes

For Examiner's use only		
Question	**Maximum Mark**	**Mark Awarded**
1.	**7**	
2.	**7**	
3.	**9**	
4.	**6**	
5.	**7**	
6.	**7**	
7.	**9**	
8.	**12**	
9.	**6**	
10.	**10**	
Total	**80**	

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ADDITIONAL MATERIALS

In addition to this examination paper you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **9** is a quality of extended response (QER) question where your writing skills will be assessed.

The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.

JUN193410UA0101

Examiner only

Answer **all** *questions.*

1. *(a)* The following table shows some information about Group 1 elements.

Metal	Melting point (°C)	Boiling point (°C)	Density (g/cm^3)	Reaction with chlorine
lithium	180	1342	0.54	reacts slowly to make a white salt
sodium	97	883	0.97	burns vigorously with a yellow flame to make a white salt
potassium	63	759	0.88	reacts violently to make a white salt
rubidium	39	688	1.53	explosive reaction
caesium	28	671	1.93	explosive reaction

(i) Describe the trend in density going down the group. [1]

....

(ii) Explain the difference in reactivity down the group in terms of electronic structure. [2]

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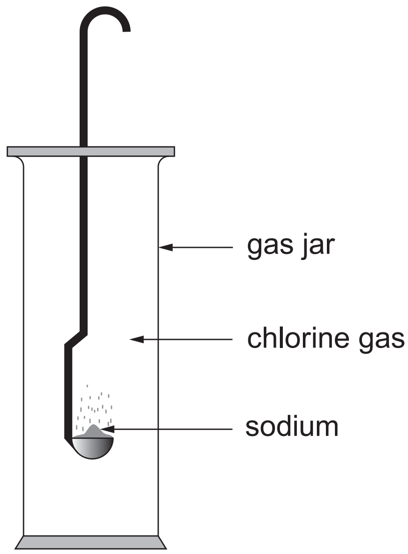
Examiner only

(b) The apparatus below can be used to demonstrate the reaction between sodium and chlorine.



(i) Apart from the use of safety goggles, state **one** safety precaution that needs to be followed when using **each** of these elements. [2]

Element	Safety precaution
sodium	....
chlorine	....

(ii) Complete and balance the symbol equation for the reaction that takes place between sodium and chlorine. [2]

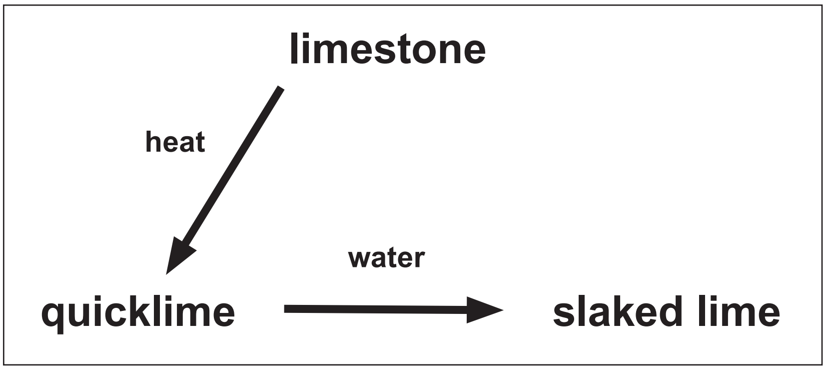
$\square$ Na + Cl_2 $\longrightarrow$ $\square$ ....

7

3410UA01 03

Examiner only

2. The following diagram shows how limestone, $CaCO_3$, can be converted into useful products.



(a) When a piece of limestone is heated strongly its mass decreases.

State the type of reaction taking place. [1]

....

(b) (i) Describe what is **seen** when limestone is heated and converted into quicklime. [1]

....

(ii) Write a balanced symbol equation for the reaction taking place. [2]

....

(c) (i) Describe what is observed when quicklime is converted into slaked lime. [1]

....

(ii) Write a balanced symbol equation for the reaction taking place. [2]

....

7

Examiner only

3. *(a)* Enzymes are biological catalysts. State what is meant by the term *catalyst*. [2]

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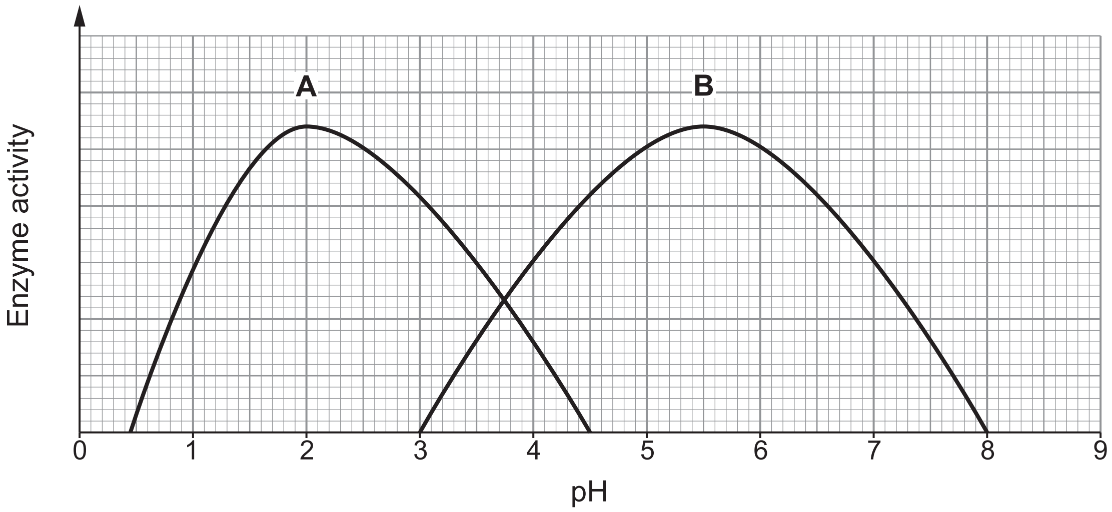
(b) The following graphs show how the activity of two enzymes, **A** and **B**, varies with pH.



(i) Use the graphs to compare the activities of the two enzymes. [2]

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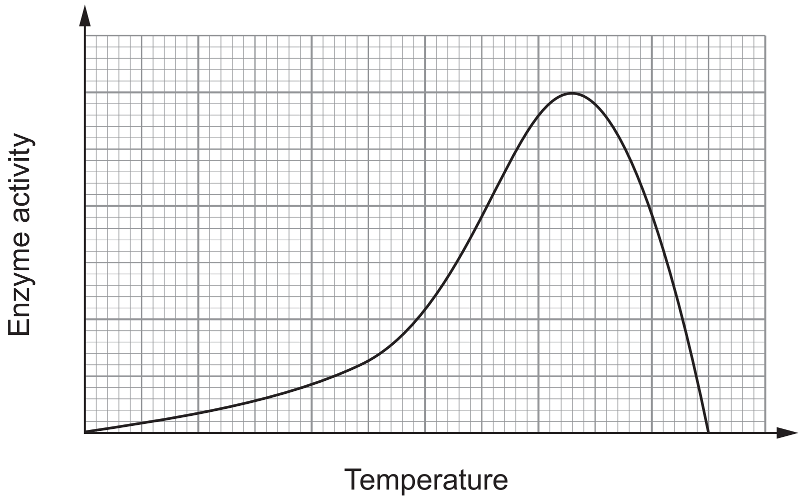
(ii) Enzyme **C** is found in saliva. It works between pH 5 and pH 9 but is best at a neutral pH. **Sketch on the grid above** how the activity varies with pH. [2]

Examiner only

(c) Temperature also affects enzyme activity as shown below.



Use the graph and your knowledge to describe how the activity of a typical enzyme changes as temperature increases. [3]

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9

Examiner only

4. Alfred Wegener proposed the theory of continental drift.

(a) State the **evidence** that Wegener used to support his theory and explain why other scientists refused to accept it. [3]

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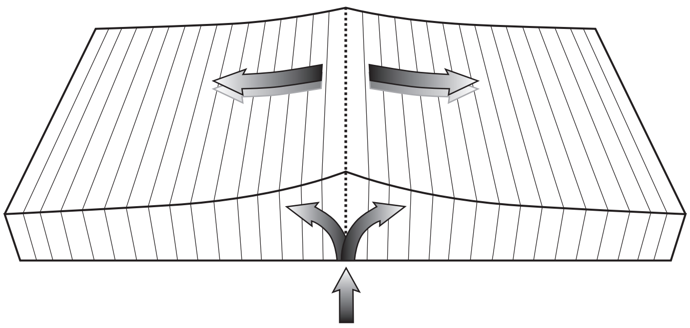
(b) In the 1960s scientists studied the ocean floor at a constructive plate boundary. Their observations are summarised in the following diagram.



(i) Describe what is happening at this boundary. [2]

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(ii) **Label the diagram** clearly to show the **oldest** rock. [1]

3410UA01 07

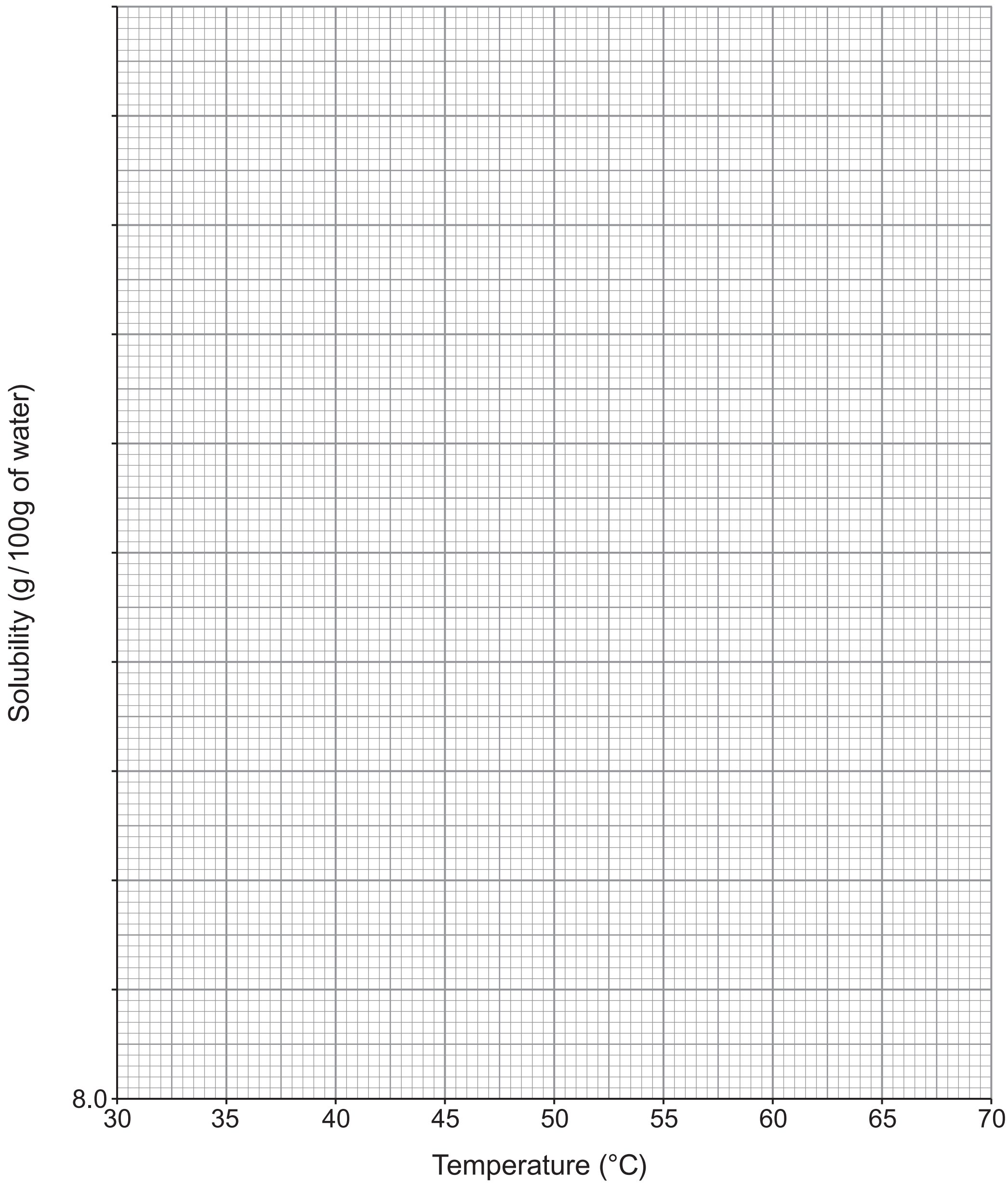
6

5. The following table shows the solubility of potassium permanganate in water at temperatures between 30 °C and 60 °C.

Temperature (°C)	Solubility (g / 100 g of water)
30	9.0
35	10.8
40	12.5
45	14.4
50	16.8
55	19.2
60	22.2

Examiner only

(a) Plot these values on the grid below. Draw a suitable line. [4]



3410UA01 09

(b) Use the graph to calculate the mass of crystals formed when a saturated solution in 500 g of water is cooled from 65 °C to 30 °C. [3]

Mass of crystals = ... g

7

6. This question is about the uses of gases present in air.

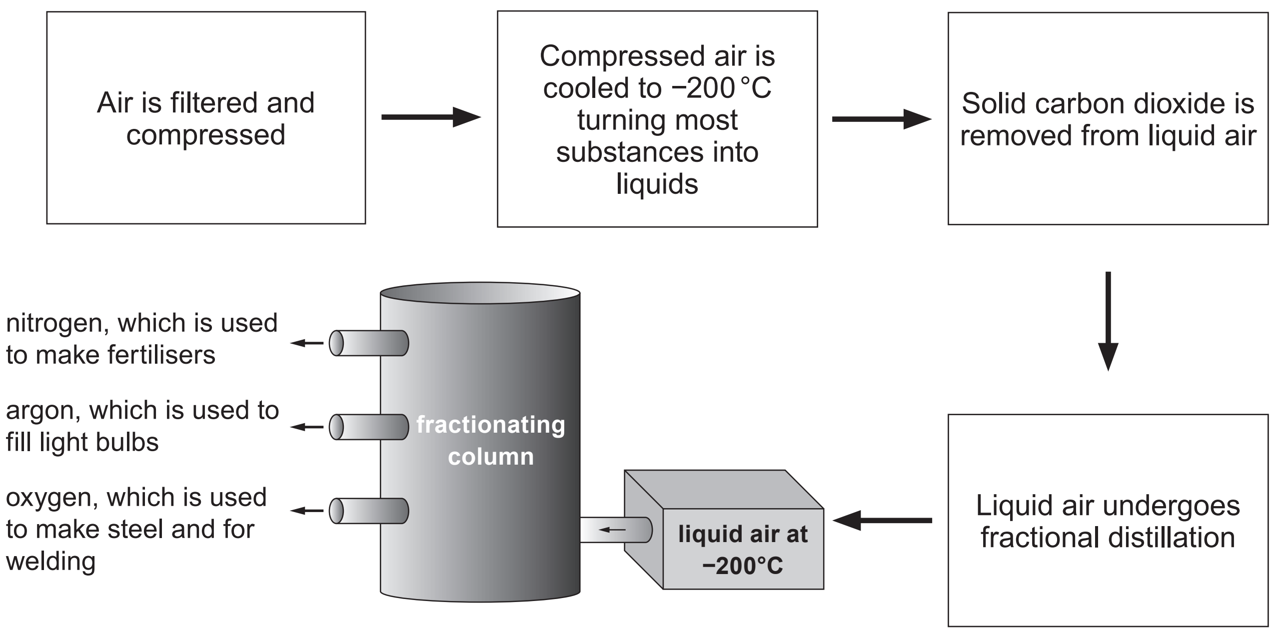
Air is a mixture of different gases. Each gas has different uses. The gases are present in different amounts as shown below.

Gas	Amount present in air	Boiling point (°C)
N_2	78.04 %	−195.8
O_2	20.95 %	−183.0
CO_2	0.03 %	−78.5
Ar	0.93 %	−189.2
Ne	18 ppm	−246.0
He	5 ppm	−268.9
Kr	1 ppm	−152.3
Xe	0.08 ppm	−107.1
H_2	0.5 ppm	−257.9
CH_4	2 ppm	−164.0
N_2O	0.5 ppm	−88.5

ppm = parts per million

Separating a complex mixture!

Air is firstly compressed and cooled which turns it into a liquid. Carbon dioxide freezes and is removed as solid dry ice. The rest of the mixture goes on to a fractionating column where it is slowly warmed allowing most of the substances to be separated.



Examiner only

(a) Tick (✓) the **main** reason why hydrogen is not separated during the fractional distillation of liquid air. [1]

hydrogen is a highly reactive gas ☐

only 0.5 ppm of hydrogen is present ☐

hydrogen does not become liquid on cooling to −200 °C ☐

hydrogen has a higher boiling point than helium ☐

(b) Tick (✓) the reason why carbon dioxide becomes a solid during the first part of the process. [1]

carbon dioxide has a boiling point above −200 °C ☐

carbon dioxide has a melting point above −200 °C ☐

carbon dioxide has a melting point below −200 °C ☐

carbon dioxide has a boiling point below −200 °C ☐

(c) Describe, in terms of boiling points, how nitrogen, argon and oxygen are separated during fractional distillation. [3]

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Examiner only

(d) The noble gases are present in very small quantities in the air. The amount of each gas produced per year is shown in the table.

Noble gas	World production (tonnes per year)
helium	28000
neon	1000
argon	700000
krypton	8
xenon	0.6

Use the data to calculate the number of tonnes of air needed to produce 700000 tonnes of argon. Give your answer in **standard form**. [2]

Mass of air needed = ... tonnes

7

Examiner only

7. *(a)* Potassium has three stable isotopes – ^{39}K, ^{40}K and ^{41}K.

(i) Compare the nuclei of each of these isotopes. [1]

....

....

....

(ii) Use the information to calculate the relative atomic mass, A_r, of potassium.

Record your answer to **three** significant figures. [3]

Isotope	Relative isotopic mass	% in sample
^{39}K	39	93.1
^{40}K	40	0.0122
^{41}K	41	6.88

$$A_r = \frac{(\text{mass } \times \text{ \% isotope 1}) + (\text{mass } \times \text{ \% isotope 2}) + (\text{mass } \times \text{ \% isotope 3})}{100}$$

A_r = ...

Examiner only

(b) Lithium lies above potassium in the Periodic Table.

(i) Give **two** similarities and **two** differences between the reactions of potassium and lithium with water. [2]

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(ii) Write a balanced symbol equation for the reaction between potassium and water. [3]

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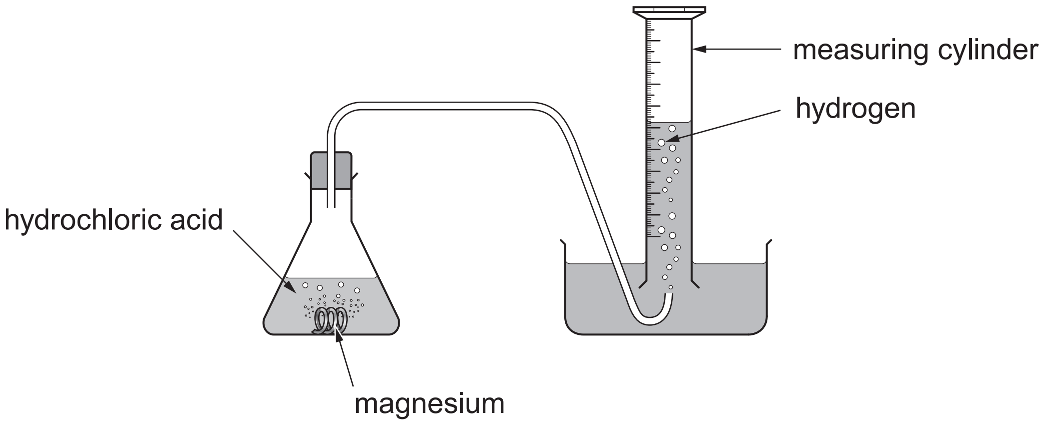
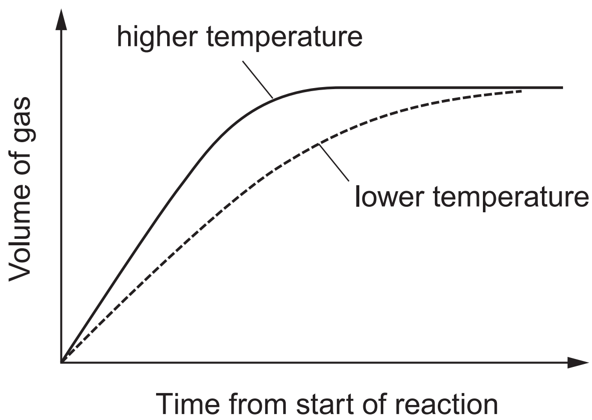
9

Examiner only

8. The following apparatus can be used to investigate the rate of the reaction between magnesium and excess dilute hydrochloric acid.



The results obtained at two different temperatures are shown below.



(a) Explain the results obtained at different temperatures in terms of particle theory. [3]

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Examiner only

(b) Explain the change in rate over time. [2]

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(c) State **one** way of improving the validity of the results obtained. Explain your answer. [2]

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Examiner only

(d) The equation for the reaction taking place is as follows.

$$Mg \ + \ 2HCl \longrightarrow MgCl_2 \ + \ H_2$$

(i) In the experiment, 0.445 g of magnesium was used.

Calculate the mass of hydrogen that will be produced during the reaction. [3]

$A_r(Mg) = 24$ $A_r(H) = 1$

Mass = ... g

(ii) It is known that one mole of gas has a volume of 24 dm^3.

Use the equation below to calculate the volume of hydrogen produced. [2]

$$\text{number of moles} = \frac{\text{volume of gas}}{24}$$

Volume = ... dm^3

12

Examiner only

9. A student is investigating the relative reactivities of the halogens using the elements and solutions of their halides. She adds each halogen to separate solutions of the other halides.

State and explain the observations made. How can the results be used to determine the relative reactivities of the halogens? Include equations in your answer. [6 QER]

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Examiner only

10. *(a)* Global warming is believed to be mainly the result of increasing levels of carbon dioxide in the atmosphere.

(i) State how the balance of carbon dioxide and oxygen is maintained in the atmosphere. Explain why levels of carbon dioxide are increasing and how this leads to global warming. [3]

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(ii) Carbon capture and storage (CCS) is one possible method of reducing global warming.

Describe briefly how CCS is carried out. [2]

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Examiner only

(b) Oxides of nitrogen are released in vehicle exhaust fumes. They can cause smog in busy city centres as well as acid rain.

(i) One oxide of nitrogen contains 30.4 g of nitrogen and 69.6 g of oxygen.

Find the simplest formula of this oxide. You **must** show your working. [3]

$A_r(N) = 14$ $A_r(O) = 16$

Simplest formula ....

(ii) The oxide with this simplest formula is found to have a relative molecular mass of 92. Find its molecular formula. [2]

Molecular formula ....

10

END OF PAPER

FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	**Formula**	**Name**	**Formula**
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		

THE PERIODIC TABLE

Group

1	2											3	4	5	6	7	0
						1 H Hydrogen 1											4 He Helium 2
7 Li Lithium 3	9 Be Beryllium 4											11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10
23 Na Sodium 11	24 Mg Magnesium 12											27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
86 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86
223 Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89															

Key

A_r — relative atomic mass Symbol Name Z — atomic number

Surname	Centre Number	Candidate Number
First name(s)		0

GCSE

3410UA0-1

Z22-3410UA0-1

FRIDAY, 17 JUNE 2022 – AFTERNOON

CHEMISTRY – Unit 1:
Chemical Substances, Reactions and Essential Resources

HIGHER TIER

1 hour 45 minutes

For Examiner's use only		
Question	**Maximum Mark**	**Mark Awarded**
1.	**5**	
2.	**9**	
3.	**6**	
4.	**9**	
5.	**5**	
6.	**8**	
7.	**5**	
8.	**7**	
9.	**11**	
10.	**9**	
11.	**6**	
Total	**80**	

3410UA01
01

ADDITIONAL MATERIALS

In addition to this examination paper you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid. You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

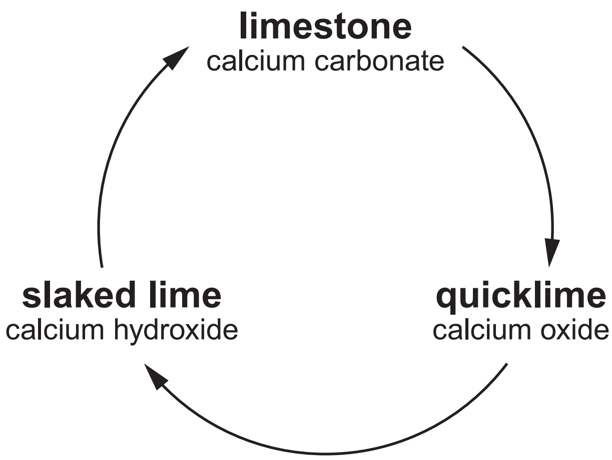
Question **9**(a) is a quality of extended response (QER) question where your writing skills will be assessed.

The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.

Examiner only

Answer **all** questions.

1. Limestone is a rock which consists mostly of calcium carbonate. The diagram shows a cycle of reactions involving limestone.



(a) (i) When limestone is heated, calcium carbonate is converted to calcium oxide and carbon dioxide.

I. State the name for this type of reaction. [1]

....

II. Write a balanced symbol equation for the reaction. [2]

.... $\longrightarrow$ .... + ...

(ii) State what must be added to calcium oxide to form calcium hydroxide. [1]

....

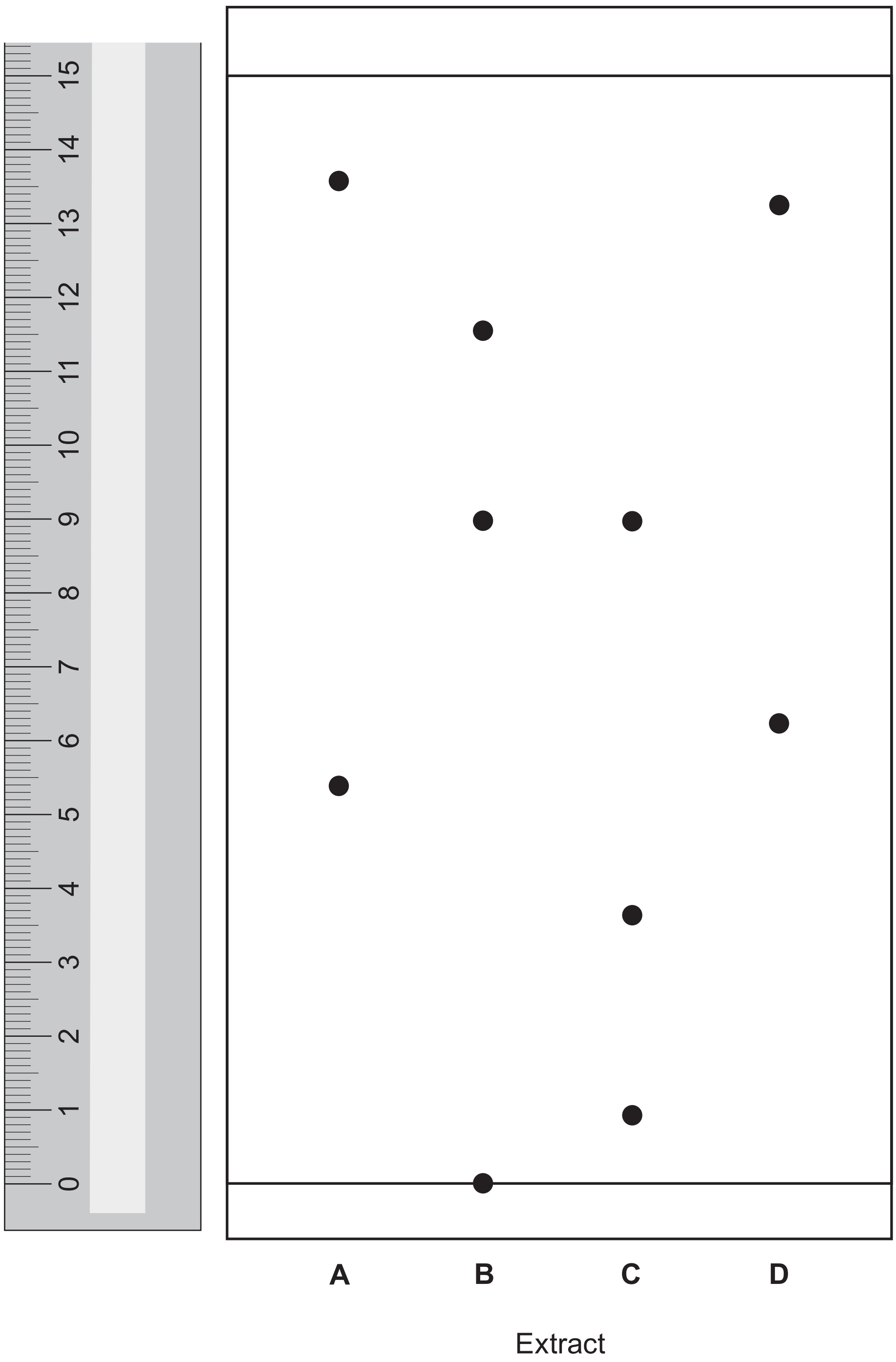
(b) Give **one** use of limestone in the construction industry. [1]

....

5

2. Paper chromatography can be used to identify plant leaf pigments. This process uses a chemical called acetone as the solvent instead of water.

The diagram shows the chromatogram of plant leaf extracts **A**, **B**, **C** and **D** in acetone.



Examiner only

(a) All of the extracts contain a mixture of pigments with different R_f values.

For which plant leaf extract, **A**, **B**, **C** or **D**, is the **highest** R_f value 0.60?

Give your reasoning. [3]

Extract ....

Reasoning ....

....

....

(b) Explain why the pigments travel different distances on the chromatogram. [2]

....

....

....

(c) One of the extracts contains a pigment which is insoluble in acetone.

State the **letter** of this extract. Explain your choice. [2]

Extract ....

Explanation ....

....

(d) The chemical formula of the solvent acetone is C_3H_6O. Calculate the percentage by mass of carbon in acetone.

The relative formula mass (M_r) of acetone is 58. [2]

$A_r(C) = 12$

Percentage = ... %

9

3410UA01 05

Examiner only

3. (a) The Earth's early atmosphere contained large amounts of water vapour and carbon dioxide. Explain why the amounts of water vapour and carbon dioxide decreased over geological time. [4]

Water vapour ....

....

....

....

Carbon dioxide ....

....

....

....

(b) State the percentages of nitrogen and oxygen in the present atmosphere. [2]

Nitrogen .... %

Oxygen .... %

6

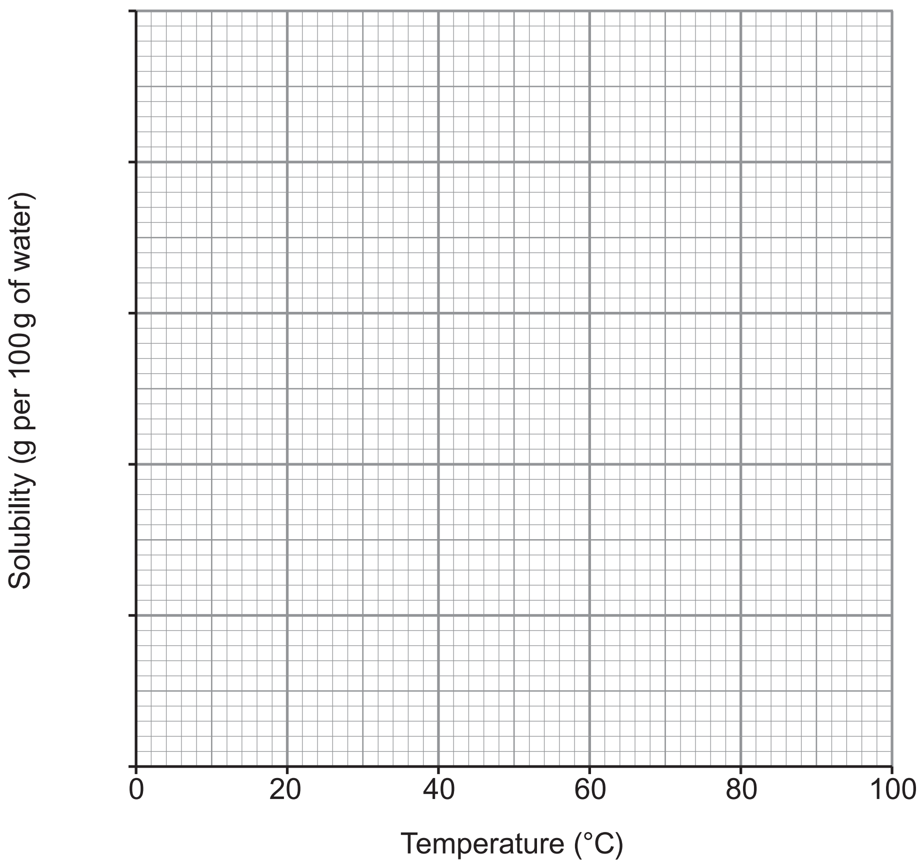
Examiner only

4. (a) The table shows the solubility of potassium nitrate in water at different temperatures.

Temperature (°C)	Solubility (g per 100 g of water)
0	13
20	32
40	64
60	110
80	169
100	246

(i) Choose a suitable scale for the *y*-axis and plot the data on the grid.

Draw a line of best fit. [4]

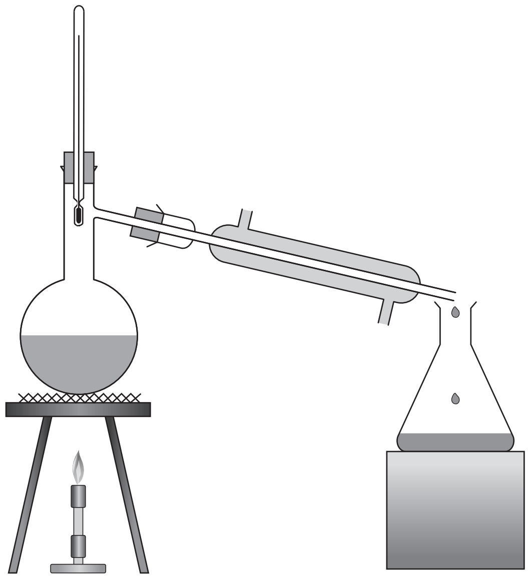


Examiner only

(ii) Calculate the mass in grams of potassium nitrate that would be crystallised if $250\,cm^3$ of a saturated solution were cooled from 80 °C to 30 °C. [3]

Mass = ... g

(b) Ethanol can be separated from water by distillation using the apparatus shown.



Explain how ethanol and water are separated by distillation. [2]

....

....

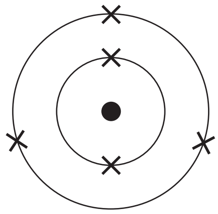
....

3410UA01
09

9

Examiner only

5. (a) The diagram shows an atom of element **X**.



Draw the electronic structure of the element directly **below** element **X** in the Periodic Table. [1]

Examiner only

(b) The table below shows information about particles **A-F**.

The letters **A-F** are **not** the chemical symbols of the particles.

Particle	Number of protons	Number of electrons	Number of neutrons
A	9	10	10
B	6	6	8
C	7	7	7
D	6	6	6
E	9	9	10
F	3	2	4

(i) Give the **letters** of **two** particles which are **isotopes** of the same element. Explain your answer. [2]

Letters .... and ....

....

....

(ii) Give the **letters** of **two** particles which are **ions**. Explain your answer. [2]

Letters .... and ....

....

....

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11

5

Examiner only

6. (a) **S** and **T** are two metal halide compounds. The ions in each compound were identified by observations from a flame test and the addition of silver nitrate solution.

(i) Complete the table. [4]

Compound	Flame test colour	Symbol of ion	Observation on adding silver nitrate solution	Symbol of ion
S	brick red	....	yellow precipitate	....
T	....	Ba^{2+}	white precipitate	....

(ii) Suggest why a silver nitrate test would not be able to distinguish clearly which halide ions are present in a **mixture** of compounds **S** and **T**. [1]

....

....

(b) Write an **ionic** equation for the formation of the precipitate in the reaction between sodium chloride solution and silver nitrate solution.

Include state symbols. [3]

.... + ... $\longrightarrow$ ....

8

7. A catalyst is a substance that causes significant changes to the rate of a chemical reaction. Catalysts interact with the reactants in a chemical reaction and eventually release the product allowing the catalyst to be recovered.

Enzymes are biological molecules which catalyse reactions in living organisms. Enzymes tend to have an optimum temperature of 37 °C.

Hydrogen peroxide is a colourless liquid with the chemical formula H_2O_2. It has many uses including as a bleach and an antiseptic. It is also found in all animals and plants. The decomposition of hydrogen peroxide happens very slowly on its own.

$$2H_2O_2(aq) \longrightarrow 2H_2O(l) + O_2(g)$$

The reaction can be catalysed by several substances including iron(III) oxide, Fe_2O_3, manganese dioxide, MnO_2, and the enzyme catalase.

Four catalysts were tested to determine which was the most efficient. An equal mass of each catalyst was added to equal volumes of hydrogen peroxide solution of equal concentration. The volume of oxygen produced in 1 minute was recorded. The experiment was repeated at different temperatures.

Temperature (°C)	Volume of oxygen produced in 1 minute (cm^3)			
	Catalyst **W**	Catalyst **X**	Catalyst **Y**	Catalyst **Z**
10	0.8	1.2	0.4	0.9
20	1.4	1.4	0.9	1.5
30	1.9	2.1	1.3	2.1
40	2.5	2.6	1.6	2.5
50	3.4	3.8	1.2	3.5
60	4.6	4.8	0.7	4.7
70	5.2	5.4	0.3	5.3

Examiner only

(a) Give the **number** of the statement which is **not** correct. [1]

1 Catalyst **W** is more efficient than **Y** but less efficient than **X** and **Z**

2 Catalyst **X** is the most efficient of the four

3 Catalyst **Y** is less efficient than all of the others

4 Catalyst **Z** is more efficient than **Y** but less efficient than **W** and **X**

....

(b) State which catalyst, **W**, **X**, **Y** or **Z**, could be catalase. Explain your answer. [3]

Catalyst ....

....

....

....

(c) Three students made statements about the properties of catalysts.

- Katrina said that a catalyst does not get used up in the reaction but makes it happen faster.
- Jo said that the reactants are used up in less time because a catalyst makes a different product.
- Rhys said that adding a catalyst allows the reaction to happen with a lower activation energy, but the same products are made.

Which of the students' statements are correct? [1]

A Jo only

B Rhys and Jo, but not Katrina

C Katrina and Rhys, but not Jo

D All three students

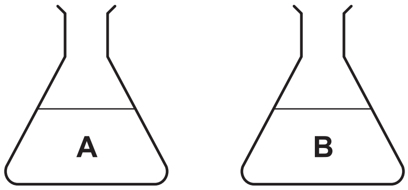
....

5

Examiner only

8. Hard water can be softened using different methods, including boiling the water and passing it through an ion exchange column.

(a) Two hard water samples, **A** and **B**, were tested for hardness, using soap solution. Both samples were tested before and after boiling, and after passing through an ion exchange column.



	Volume of soap solution needed to produce a lather (cm^3)		
Water sample	Before boiling	After boiling	After ion exchange
A	15	15	2
B	17	9	2

State the type (or types) of hard water that each sample contains. Give your reasoning. [4]

A ....

....

....

B ....

....

....

....

....

Examiner only

(b) Sodium carbonate (washing soda) can also be used to soften hard water. It reacts with the magnesium compounds in the water.

Write a balanced symbol equation for the reaction between sodium carbonate and magnesium chloride, to form sodium chloride and magnesium carbonate. [3]

....

7

Examiner only

9. (a) Give the observations made during the reactions of lithium, sodium and potassium with water. Explain the trend in reactivity in terms of electronic structure. [6 QER]

Examiner only

(b) Potassium reacts with oxygen to produce potassium oxide.

$$4K + O_2 \longrightarrow 2K_2O$$

(i) Calculate the maximum mass of potassium oxide that could be produced when 15.6 g of potassium reacts completely with oxygen. [3]

$A_r(K) = 39$ $A_r(O) = 16$

Mass = ... g

(ii) Another reaction requires 0.050 mol of oxygen gas. Calculate the number of molecules in 0.050 mol of oxygen gas. Give your answer in **standard form**. [2]

Avogadro's number = 6.0×10^{23}

Number of molecules = ...

11

10. (a) Tables **A**, **B**, **C** and **D** show the results recorded by four different groups in a class investigating the reactivity of Group 7 elements (the halogens).

Each group added iodine, bromine and chlorine to solutions of sodium halides.

A tick (✓) indicates when a reaction took place and a cross (×) indicates no reaction. Only **one** table shows the expected results for this experiment.

Table **A**

	sodium chloride	sodium bromide	sodium iodide
bromine	✓	✓	×
chlorine	✓	×	✓
iodine	✓	✓	✓

Table **B**

	sodium chloride	sodium bromide	sodium iodide
bromine	✓	✓	✓
chlorine	×	×	✓
iodine	×	✓	✓

Table **C**

	sodium chloride	sodium bromide	sodium iodide
bromine	×	×	✓
chlorine	×	✓	✓
iodine	×	×	×

Table **D**

	sodium chloride	sodium bromide	sodium iodide
bromine	×	✓	×
chlorine	✓	×	✓
iodine	×	✓	✓

Examiner only

(i) Give the **letter** of the table which shows the expected results for this experiment. Explain why these are the expected results. [3]

Letter ....

....

....

....

(ii) Write a balanced symbol equation for the reaction between chlorine, Cl_2, and sodium iodide solution. [3]

....

(b) A sample of iron bromide with a mass of 37.0 g contains 7.0 g of iron. Find the simplest formula of the iron bromide.

You **must** show your working. [3]

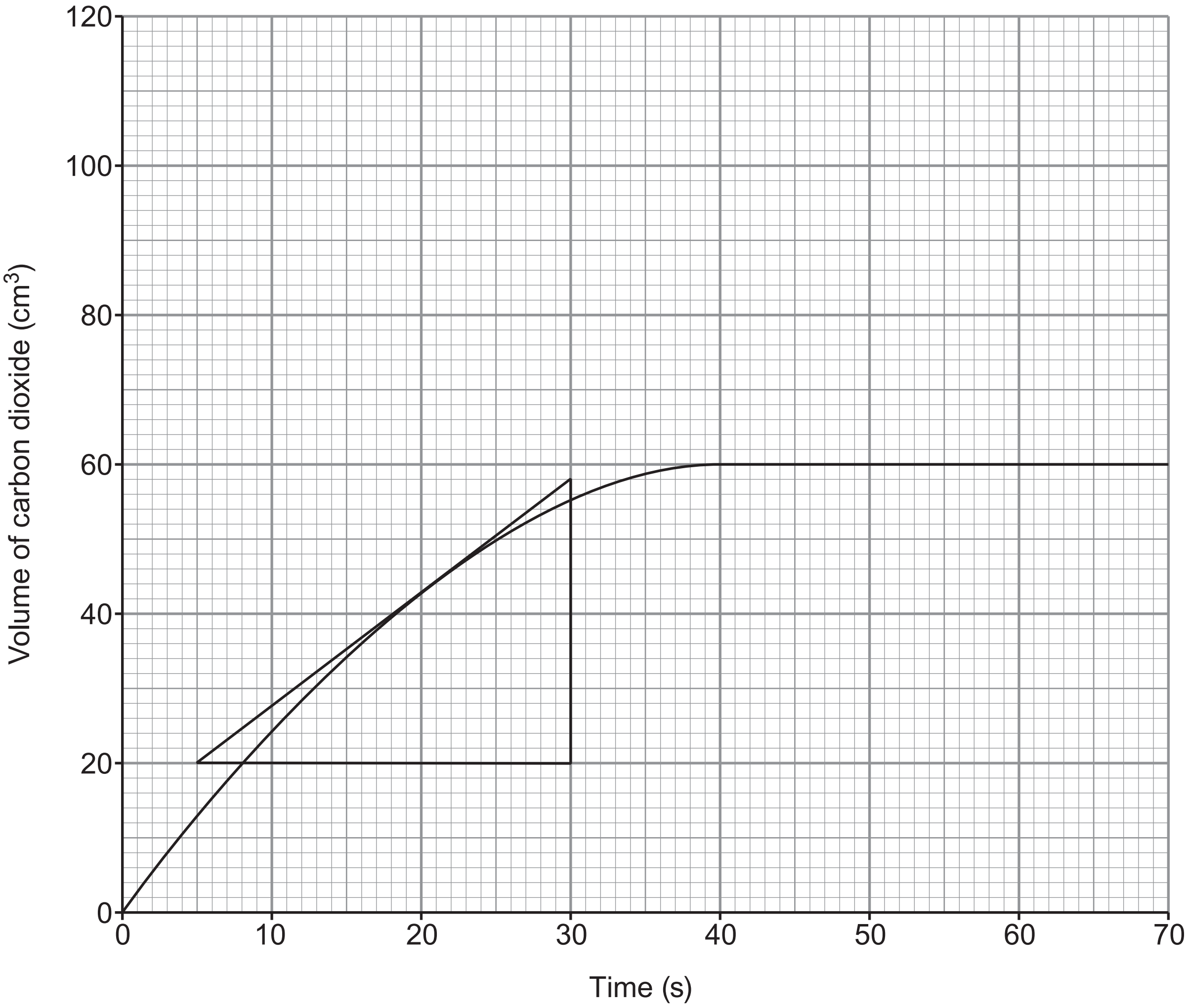
A_r(Fe) = 56 A_r(Br) = 80

Simplest formula ....

9

Examiner only

11. The graph shows the volume of gas produced in a reaction between 0.40 g of calcium carbonate powder and excess 1 mol / dm^3 hydrochloric acid.



(a) Use the tangent drawn to calculate the rate of the reaction at 20 s. [2]

Rate = ... cm^3 / s

 (3410UA0-1)

Examiner only

(b) (i) On the same grid, sketch the graph that would be seen if the reaction were repeated with 0.60 g of calcium carbonate powder and acid of the same concentration and still in excess. [2]

(ii) Explain the graph that you have sketched using your knowledge of particle theory. [2]

....

....

....

....

....

....

6

END OF PAPER

FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	**Formula**	**Name**	**Formula**
aluminium	Al^{3+}	bromide	Br^{-}
ammonium	NH_4^{+}	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^{-}
calcium	Ca^{2+}	fluoride	F^{-}
copper(II)	Cu^{2+}	hydroxide	OH^{-}
hydrogen	H^{+}	iodide	I^{-}
iron(II)	Fe^{2+}	nitrate	NO_3^{-}
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^{+}	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^{+}		
silver	Ag^{+}		
sodium	Na^{+}		
zinc	Zn^{2+}		

THE PERIODIC TABLE

Group

1	2											3	4	5	6	7	0
						1 H Hydrogen 1											4 He Helium 2
7 Li Lithium 3	9 Be Beryllium 4											11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10
23 Na Sodium 11	24 Mg Magnesium 12											27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
86 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86
223 Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89															

Key

A_r — relative atomic mass Symbol Name Z — atomic number

Surname	Centre Number	Candidate Number
First name(s)		0

GCSE

3410UA0-1

S23-3410UA0-1

FRIDAY, 16 JUNE 2023 – MORNING

CHEMISTRY – Unit 1:
Chemical Substances, Reactions and Essential Resources
HIGHER TIER

1 hour 45 minutes

For Examiner's use only		
Question	**Maximum Mark**	**Mark Awarded**
1.	**9**	
2.	**11**	
3.	**9**	
4.	**8**	
5.	**5**	
6.	**9**	
7.	**12**	
8.	**9**	
9.	**8**	
Total	**80**	

3410UA01
01

ADDITIONAL MATERIALS

In addition to this examination paper you will require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet.

If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

The assessment of the quality of extended response (QER) will take place in Question **6**(a).

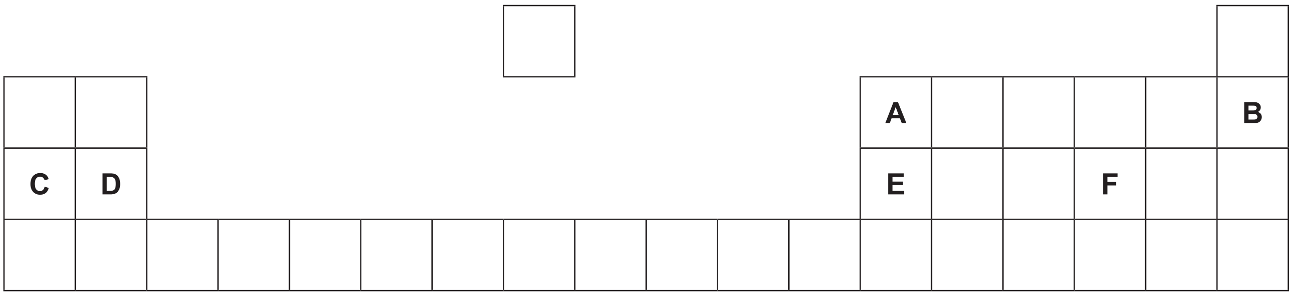
The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.

JUN233410UA0101

Answer **all** questions.

1. (a) The following diagram shows an outline of part of the Periodic Table.

The letters shown are NOT the chemical symbols of the elements.

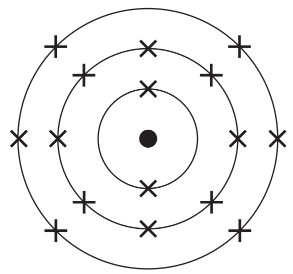


Choose **letters** from the diagram to complete the table below. [4]

	Letter
The element in Group 3 and Period 2	…………………………….
The element which has 10 protons in its nucleus	…………………………….
The element with the electronic structure 2,8,6	…………………………….
The element which forms a 2+ ion	…………………………….

Examiner only

(b) The diagram below shows the electronic structure of an element in the Periodic Table.



In the space below, draw a diagram to show the electronic structure of the element which lies directly **above** it. [1]

3410UA01 03

(c) The table shows information about atoms **X**, **Y** and **Z**.

Atom	Symbol	Number of protons	Number of neutrons	Number of electrons
X	$^{31}_{15}\mathbf{X}$	....	16	15
Y	$^{39}_{19}\mathbf{Y}$	19	....	19
Z	$^{40}_{19}\mathbf{Z}$	19	21	....

(i) Complete the table. [3]

(ii) Underline the term used to describe atoms **Y** and **Z**. [1]

ions **inert** **insoluble** **isotopes**

9

Examiner only

2. (a) The table shows information about some Group 1 elements.

Element	Relative atomic mass	Number of electrons in the outer shell	Melting point (°C)	Boiling point (°C)	Density (g/cm^3)
lithium	7	1	180	1342	0.53
sodium	23	1	98	883	0.97
potassium	39	1	63	759	0.89
rubidium	85	1	39	688	1.53
caesium	134	1	29	671	1.93

Use the information in the table to answer parts (i) and (ii).

(i) State the information which explains why the elements have similar chemical properties. [1]

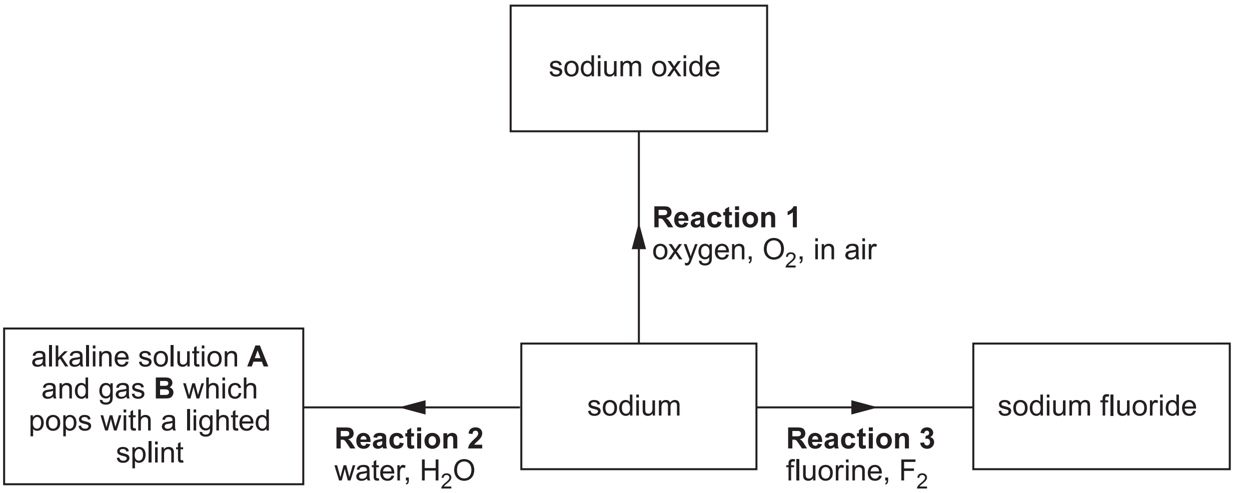
....

(ii) State which **property** has a value which does **not** fit the trend down the group. [1]

....

Examiner only

(b) The flow diagram shows some reactions of sodium.



(i) State how **Reaction 1** is prevented when storing sodium in the laboratory. [1]

....

(ii) Give the names of alkaline solution **A** and gas **B**. [2]

.... and ....

(iii) Name the Group 1 metal which would react **least** violently with water. [1]

....

(iv) Complete the symbol equation for **Reaction 3**. [1]

$$2Na + F_2 \longrightarrow 2\ \ldots\ldots\ldots\ldots$$

3410UA01 05

Examiner only

(c) Sodium fluoride is added to some UK public water supplies to reduce tooth decay in children.

In America sodium hexafluorosilicate, Na_2SiF_6, is more commonly used. The relative formula mass of sodium hexafluorosilicate is 188.

(i) Calculate the percentage of fluorine in sodium hexafluorosilicate. [2]

$A_r(F) = 19 \qquad M_r(Na_2SiF_6) = 188$

Percentage = ... %

(ii) State an **ethical** reason why some people oppose the fluoridation of water supplies. [1]

....

....

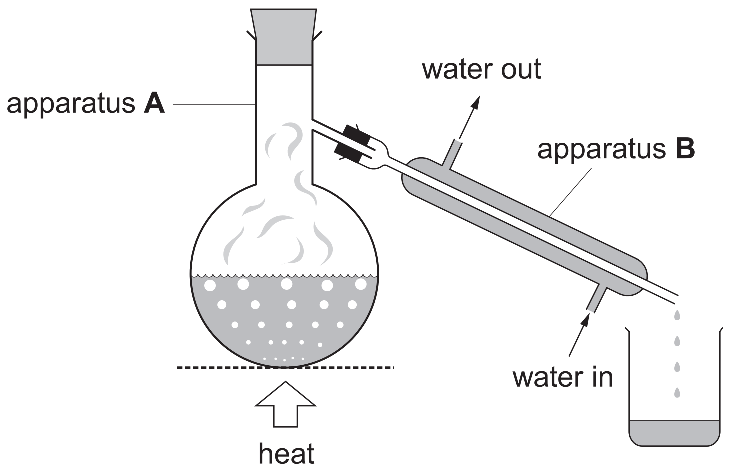
(iii) Apart from water supplies, state the most commonly used source of fluoride to reduce tooth decay. [1]

....

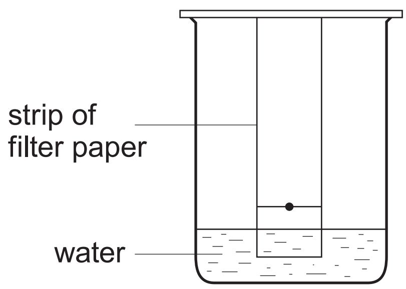
11

Examiner only

3. (a) The diagrams show two methods used to separate mixtures.



Method **1**



Method **2**

(i) I. Name the changes of state happening in apparatus **A** and apparatus **B** when water is separated from salt in Method **1**. [2]

Apparatus **A** ....

Apparatus **B** ....

II. Name this method of separation. [1]

....

(ii) Explain how an orange dye is separated into a red spot and a yellow spot on the filter paper in Method **2**. [2]

....

....

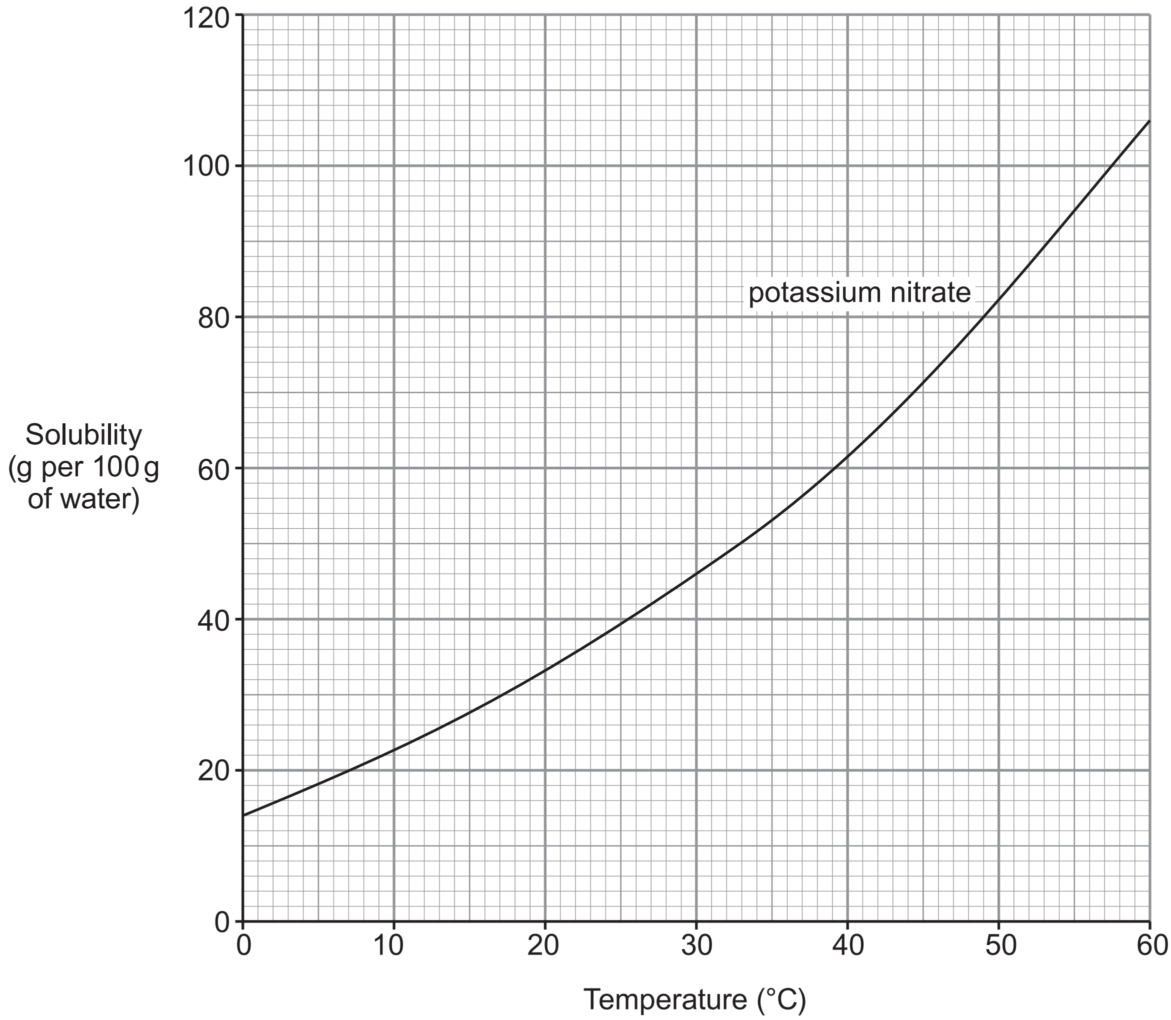
....

....

3410UA01
07

Examiner only

(b) The graph below shows the solubility of potassium nitrate at different temperatures.



Use the information in the graph to answer parts (i) and (ii).

(i) 60 g of potassium nitrate was added to 100 g of water at 30 °C. After stirring the mixture some of the potassium nitrate did not dissolve.

Calculate the mass of potassium nitrate which did **not** dissolve. [1]

Mass = ... g

Examiner only

(ii) On cooling a saturated solution of potassium nitrate containing 100 g of water from 55 °C to a lower temperature, 36 g of solid was formed.

Determine the temperature to which the solution was cooled.

Show your working. [3]

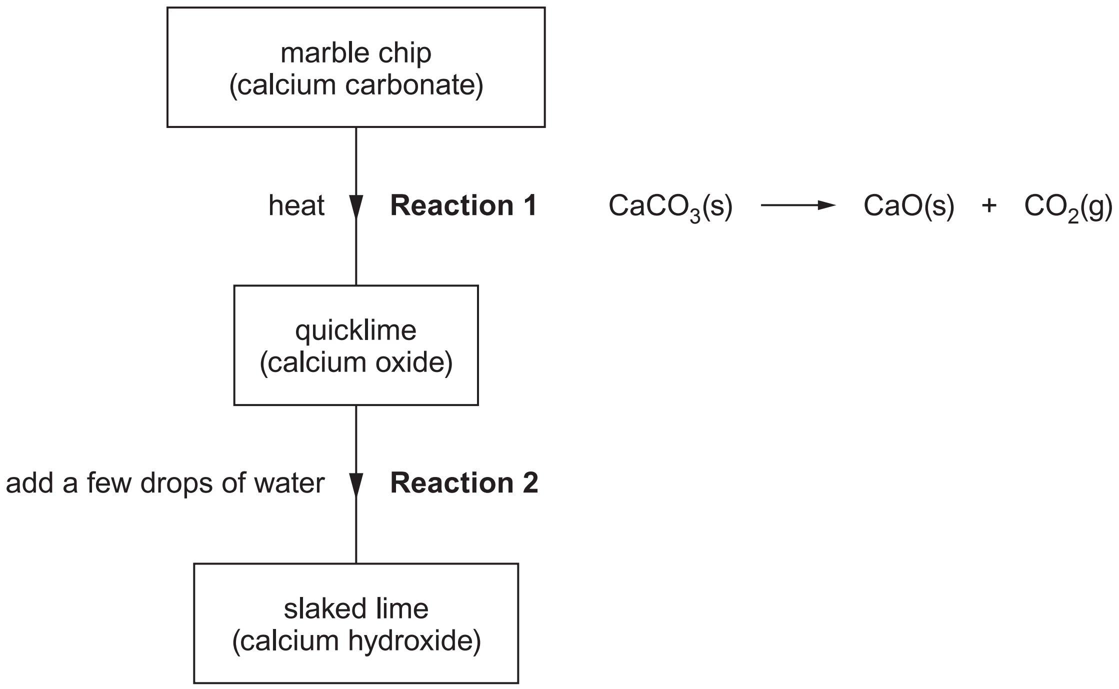
Temperature = ... °C

9

3410UA01 09

Examiner only

4. (a) The flow diagram shows the reactions necessary to make slaked lime (calcium hydroxide) from a marble chip (calcium carbonate).



(i) In **Reaction 1**, one compound breaks down when heated to form two products.

Complete the name of this type of reaction. [1]

thermal ....

(ii) Give **two** observations you would expect to make as **Reaction 2** happens. [2]

1 ....

2 ....

(iii) Complete the symbol equation for the reaction between calcium oxide and water in **Reaction 2**. [1]

$\ldots\ldots\ldots\ldots\ldots\ldots + H_2O \longrightarrow \ldots\ldots\ldots\ldots\ldots\ldots$

 (3410UA0-1)

Examiner only

(b) (i) The relative formula mass of calcium hydroxide is 74.

Calculate the number of moles of calcium hydroxide in 2.96 g. [2]

Number of moles = ... mol

(ii) Explain why calcium hydroxide is used to treat some soils. [2]

....

....

....

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Examiner only

5. Temporary hard water is caused by the presence of dissolved calcium hydrogencarbonate, $Ca(HCO_3)_2$.

The table shows three different methods of removing temporary hardness from water.

Method	Product(s) in softened water
1. Adding sodium carbonate	insoluble calcium carbonate dissolved sodium hydrogencarbonate
2. Boiling	insoluble calcium carbonate carbon dioxide
3. Passing through an ion exchange column containing Na^+ ions	dissolved sodium hydrogencarbonate

(a) Complete the symbol equation for the reaction taking place in Method **1**. [1]

$$Ca(HCO_3)_2 \ + \ \ldots\ldots\ldots\ldots\ldots\ldots \longrightarrow CaCO_3 \ + \ 2NaHCO_3$$

(b) Underline the ratio of calcium ions to sodium ions exchanged in Method **3**. [1]

2:1 **1:1** **1:2** **2:2**

(c) Give the number of the method which does **not** form limescale. Give the reason for your answer. [2]

Number ....

Reason ....

....

(d) Tick (✓) the box next to the name of a compound that causes permanent hardness when dissolved in water. [1]

calcium sulfate ☐

potassium sulfate ☐

magnesium hydrogencarbonate ☐

sodium sulfate ☐

5

Examiner only

6. (a) The table shows the approximate percentages of some gases found in the Earth's **original** atmosphere and the Earth's atmosphere **today**.

Gas	Approximate percentage (%) of gases in the Earth's atmosphere	
	Original	**Today**
carbon dioxide	75	0.04
water vapour	25	variable, between 1-2
oxygen	0	21

Explain how the changes to these gases have taken place over geological time. [6 QER]

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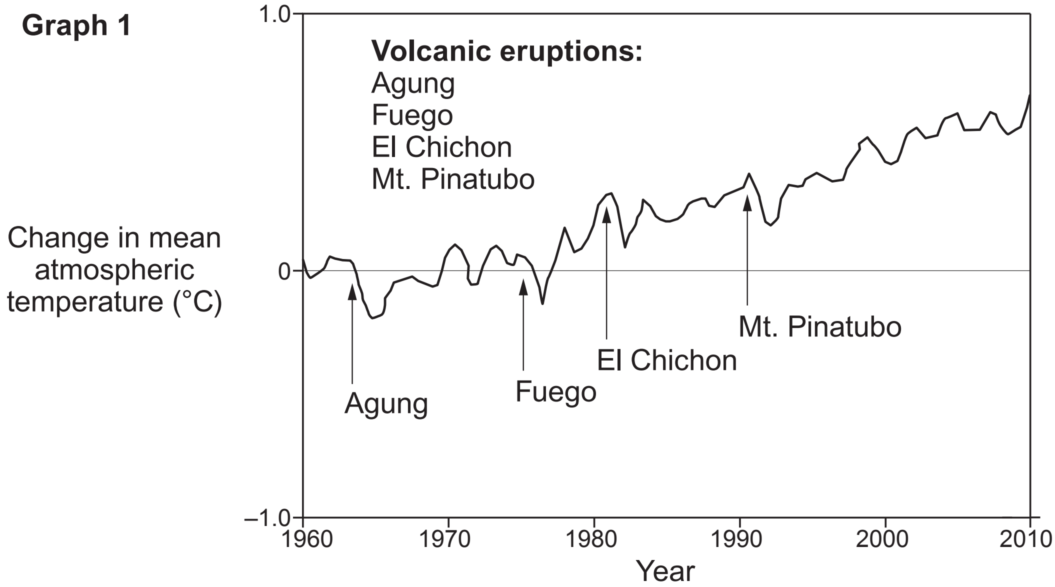
(b) Use the information on this page to answer parts (i)–(iii).

Do Volcanoes Cause Global Warming?

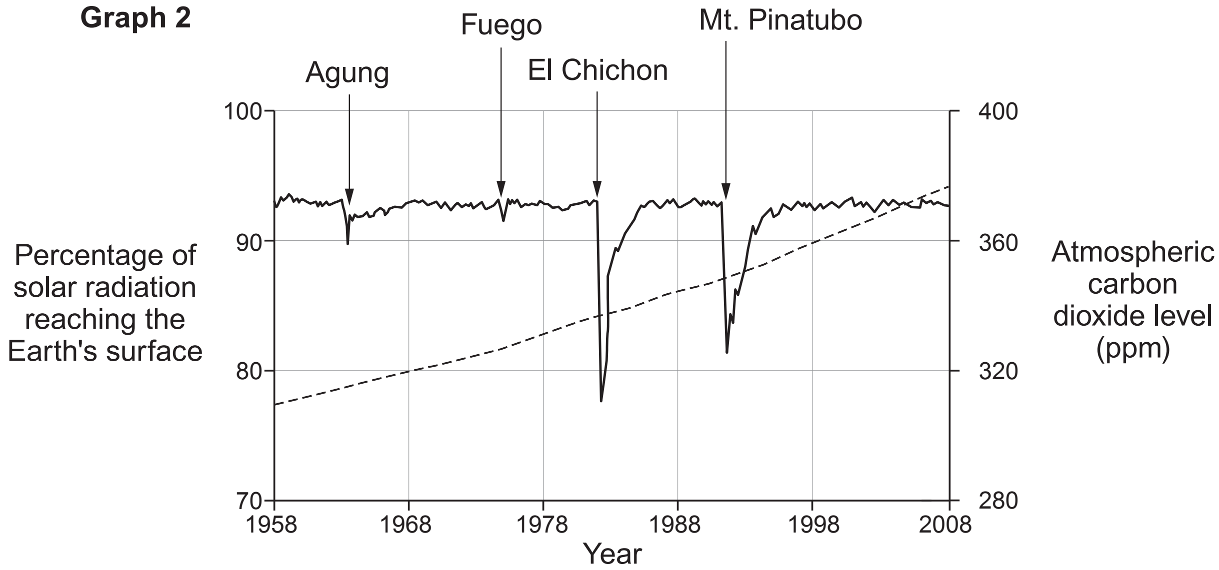
Volcanoes release vast amounts of greenhouse gases, such as carbon dioxide, which contribute to global warming. **Graph 1** shows the effect of four volcanic eruptions on the mean atmospheric temperature.

Erupting volcanoes also emit large quantities of sulfur dioxide into the atmosphere. Unlike carbon dioxide, sulfur dioxide increases the reflection of radiation from the Sun back into space, which cools the Earth's atmosphere. **Graph 2** shows the effect of the same four large volcanic eruptions on the percentage of solar radiation reaching the Earth's surface. The graph also shows the change in carbon dioxide levels in the atmosphere over the same time period.

Graph 1



Graph 2



Examiner only

(i) Tick (✓) the box next to the statement which best describes the effect of volcanic eruptions on the overall level of carbon dioxide in the atmosphere. [1]

No significant impact on the overall level of carbon dioxide ☐

A significant increase in the level of carbon dioxide ☐

A significant decrease in the level of carbon dioxide ☐

(ii) Tick (✓) the box next to the statement which best describes the effect that volcanic eruptions have on the mean atmospheric temperature. [1]

Mean atmospheric temperature decreases ☐

Mean atmospheric temperature increases ☐

No effect on the mean atmospheric temperature ☐

(iii) Tick (✓) the box next to the statement which best explains the change in solar radiation reaching the Earth's surface after volcanic eruptions. [1]

Solar radiation decreases because it is reflected by sulfur dioxide ☐

Solar radiation increases because it is absorbed by carbon dioxide ☐

Solar radiation increases because it is absorbed by carbon dioxide and sulfur dioxide ☐

Solar radiation decreases because it reacts with sulfur dioxide forming sulfuric acid ☐

9

Examiner only

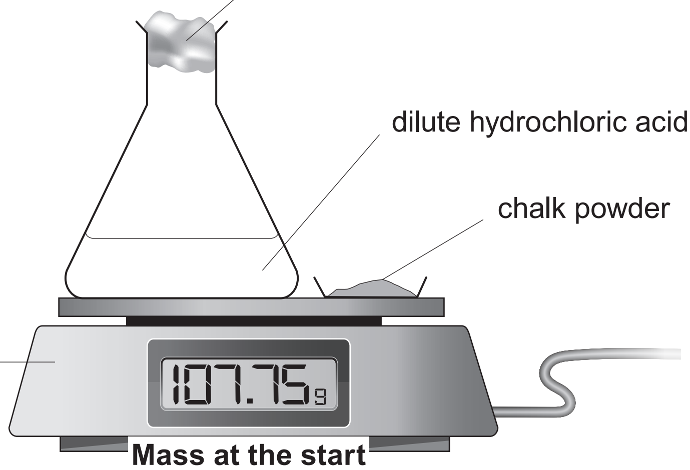
7. Chalk is a form of calcium carbonate, $CaCO_3$.

A student carried out an experiment to investigate the rate of reaction between **powdered** chalk and **excess** dilute hydrochloric acid at 20 °C.

Chalk reacts with dilute hydrochloric acid forming carbon dioxide gas.

$$CaCO_3(s) + 2HCl(aq) \longrightarrow CaCl_2(aq) + CO_2(g) + H_2O(l)$$

The diagram shows the apparatus used.



(a) State the purpose of the cotton wool. [1]

....

(b) The student recorded the mass of the flask and contents every 10 seconds for 70 seconds.

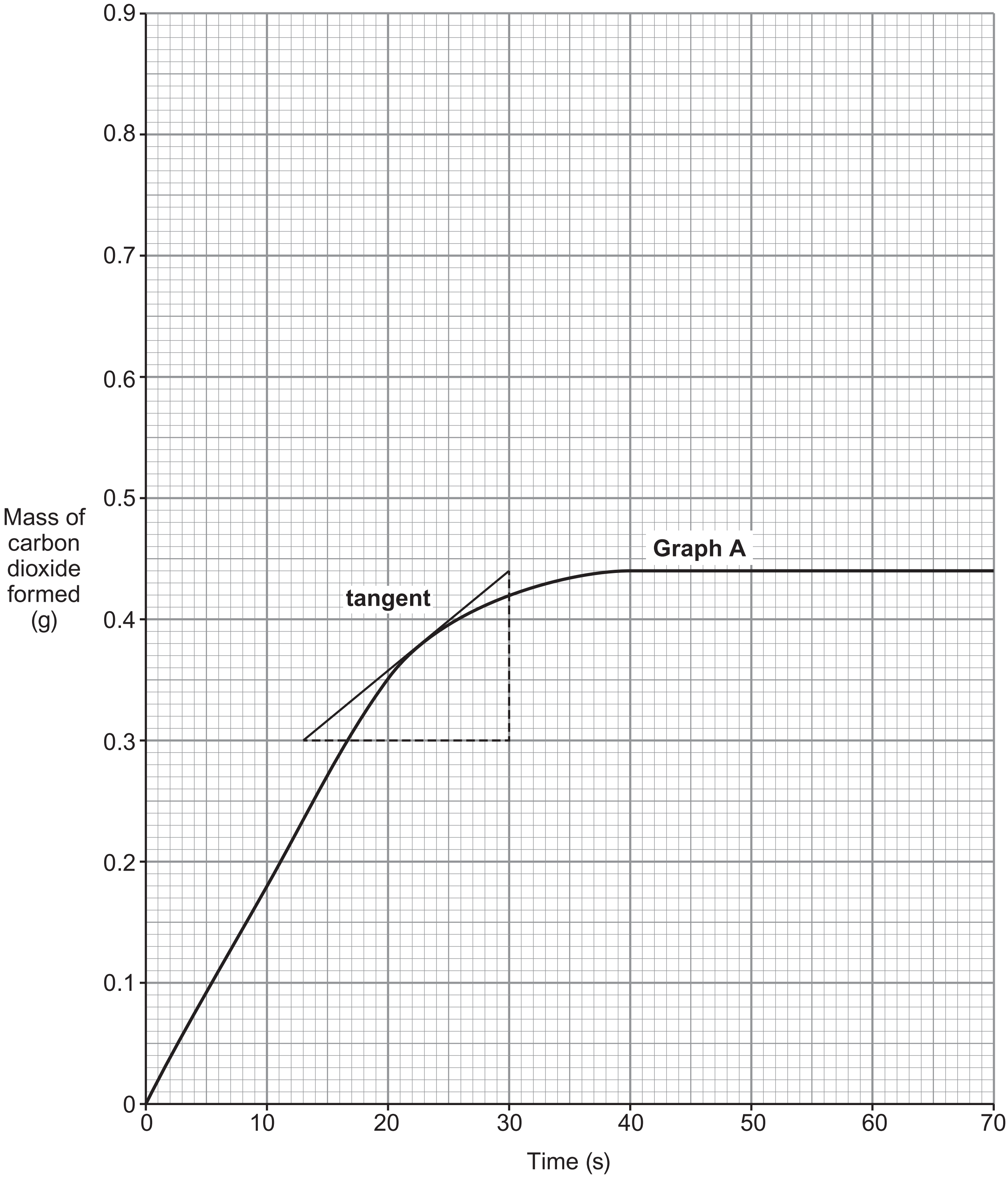
Describe what the student must do to calculate the mass of carbon dioxide formed in the first 10 seconds. [1]

....

....

(c) Graph **A** shows the mass of carbon dioxide formed against time.

A tangent to the graph has been drawn at 23 s.



Examiner only

Use the tangent and the equation below to calculate the rate of reaction at 23 s.

Show your working. Give your answer to **two** significant figures. [3]

$$\text{rate} = \frac{\text{change in mass}}{\text{change in time}}$$

Rate = ... g/s

(d) The student repeated the experiment using the **same** mass of chalk powder as the original experiment but a **different concentration** of acid. The table shows the mass of carbon dioxide formed.

Time (s)	0	10	20	30	40	50	60	70
Mass of carbon dioxide formed (g)	0.00	0.11	0.20	0.28	0.35	0.40	0.43	0.44

(i) Plot the results from the table on the grid opposite and draw a suitable line.

Label this graph **B**. [3]

(ii) Use particle theory to explain why the rate of this reaction is lower than that in the original experiment. [3]

....

....

....

....

....

(e) Sketch on the grid the curve you would expect if the experiment was repeated using the **original** acid and **twice** the mass of chalk. Assume that the acid is still in excess.

Label this graph **C**. [1]

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Examiner only

8. (a) The table shows information about some Group 7 elements.

Element	Radius of the atom (nm)	Number of electrons in outer shell
chlorine	0.099	7
bromine	0.114	7
iodine	0.133	7

(i) State how the radius of the atom changes down Group 7. [1]

....

(ii) Explain in terms of electronic structure why the reactivity of the elements decreases down Group 7. [2]

....

....

....

(b) Sea water contains sodium bromide, which is the raw material for the production of bromine.

Between 1953 and 2004 bromine, Br_2, was manufactured from sea water at a chemical plant in Amlwch, Anglesey.

Sea water was treated with chlorine, which converts sodium bromide into bromine.

Complete the balanced equation for the reaction between chlorine and sodium bromide. [3]

$$\ldots\ldots\ldots\ldots + 2NaBr \longrightarrow Br_2 + \boxed{}\ \ldots\ldots\ldots\ldots$$

Examiner only

(c) 4.47 g of copper bromide was found to contain 1.27 g of copper.

(i) Calculate the mass of bromine in 4.47 g of the copper bromide. [1]

Mass = ... g

(ii) Calculate the simplest formula of the copper bromide.

You **must** show your working. [2]

$A_r(\text{Br}) = 80$ $A_r(\text{Cu}) = 63.5$

Simplest formula ....

9

Examiner only

9. (a) A student tested compounds **A**, **B** and **C** to identify them.

Her observations are recorded in the tables below.

Compound **A**	Observation	Ion present
Flame test	apple-green flame	barium
Add silver nitrate solution	white precipitate	chloride

Compound **B**	Observation	Ion present
Flame test	lilac flame	
Add silver nitrate solution	cream precipitate	

Compound **C**	Observation	Ion present
Flame test	brick-red flame	
Add silver nitrate solution	yellow precipitate	

(i) Use your knowledge of the tests for ions to complete the tables for compounds **B** and **C**. [3]

(ii) Write the chemical formula for compound **A**. [1]

....

Examiner only

(b) When silver nitrate solution reacts with sodium chloride solution to produce a white precipitate the following reaction occurs.

$$AgNO_3(aq) + NaCl(aq) \longrightarrow NaNO_3(aq) + AgCl(s)$$

Put a tick (✓) in the box which shows the ionic equation for the formation of the precipitate. [1]

$Na^+(aq) + NO_3^-(aq) \longrightarrow AgCl(s)$	☐
$Ag^+(aq) + NO_3^-(aq) + Na^+(aq) + Cl^-(aq) \longrightarrow NaNO_3(s) + AgCl(s)$	☐
$Ag^+(aq) + NO_3^-(aq) \longrightarrow AgNO_3(s)$	☐
$Ag^+(aq) + Cl^-(aq) \longrightarrow AgCl(s)$	☐
$Ag^+(aq) + Cl^-(aq) \longrightarrow NaNO_3(s)$	☐

Examiner only

(c) When silver carbonate is heated it breaks down to give silver metal and carbon dioxide and oxygen gas.

$$2Ag_2CO_3 \longrightarrow 4Ag + 2CO_2 + O_2$$

Calculate the mass of silver that would be produced from heating 13.8 g of silver carbonate. [3]

$M_r(Ag_2CO_3) = 276 \qquad A_r(Ag) = 108$

Mass = ... g

8

END OF PAPER

FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	**Formula**	**Name**	**Formula**
aluminium	Al^{3+}	bromide	Br^{-}
ammonium	NH_4^{+}	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^{-}
calcium	Ca^{2+}	fluoride	F^{-}
copper(II)	Cu^{2+}	hydroxide	OH^{-}
hydrogen	H^{+}	iodide	I^{-}
iron(II)	Fe^{2+}	nitrate	NO_3^{-}
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^{+}	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^{+}		
silver	Ag^{+}		
sodium	Na^{+}		
zinc	Zn^{2+}		

THE PERIODIC TABLE

1	2					Group						3	4	5	6	7	0
						1 H Hydrogen 1											4 He Helium 2
7 Li Lithium 3	9 Be Beryllium 4											11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10
23 Na Sodium 11	24 Mg Magnesium 12											27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
86 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86
223 Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89															

Key

A_r — relative atomic mass Symbol Name Z — atomic number